

RAINFALL VARIABILITY IN INDIA: A STATISTICAL ANALYSIS



RAMANUJAN COLLEGE, UNIVERSITY OF DELHI

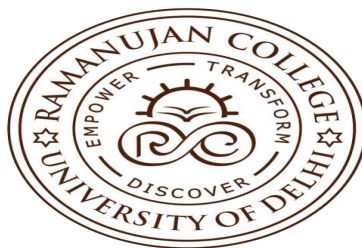
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**PROJECT MENTOR:
DR. SACHIN TOMER**

CANDIDATES:
AARADHYA ARNEJA
ADITI AGARWAL
ADITYA KUMAR
ADITYA JAISWAL
MEHAK SINGAL
YUVRAJ SINGH

**DEPARTMENT OF STATISTICS
RAMANUJAN COLLEGE, UNIVERSITY OF DELHI
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RAMANUJAN COLLEGE
University of Delhi
NAAC Grade A++ with
CGPA 3.71



रामानुजन महाविद्यालय
दिल्ली विश्वविद्यालय
सीजीपीए 3.71 . के साथ
एनएएसी ग्रेड ए++

CERTIFICATION

This is to certify that Aaradhya Arneja, Aditi Agarwal, Aditya Kumar, Aditya Jaiswal, Mehak Singal, and Yuvraj Singh, students from Ramonujan College, University of Delhi, in the batch of 2021-2024, during their sixth semester (third year), have successfully submitted a project on the topic "Rainfall Variability in India: A Statistical Analysis" to Dr. Sachin Tomer in May 2024.

I commend their diligence and dedication to academic inquiry and congratulate them on the successful completion of their project.

Dr. Sachin Tomer

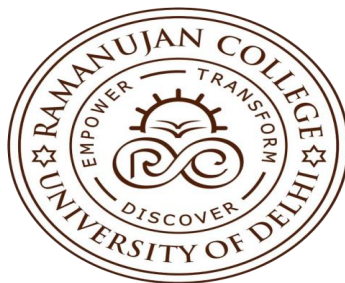
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ग्रेड ए++

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We, Aaradhya Arneja, Aditi Agarwal, Aditya Kumar, Aditya Jaiswal, Mehak Singal, and Yuvraj Singh, students of Ramanujan College, University of Delhi, batch of 2021-2024, express our sincere gratitude upon the completion of our project titled "Rainfall Variability in India: A Statistical Analysis," submitted in May 2024 during our sixth semester.

We extend our heartfelt thanks to Ramanujan College, our Principal, the college library, and the Faculty of Department of Statistics for their invaluable support. Special thanks to Dr. Sachin Tomer, our mentor, whose expert guidance was crucial to the success of this project.

We appreciate the dedication and hard work of our team, which was essential in achieving this academic accomplishment.

PREFACE

Introduction and Overview

In chapter one, we'll be delving into the captivating world of rainfall patterns and analysis. In India, many studies have looked at rainfall patterns and their effects on different areas. They used different methods to understand extreme rainfall events and how they impact agriculture, water resources, and infrastructure. For example, studies in Maharashtra and Gujarat tried to find the best ways to predict extreme rainfall and fill in missing data. In Roorkee and Uttarakhand, they focused on choosing the right methods to predict maximum rainfall, which is important for managing water and planning infrastructure.

Other research in places like Chhattisgarh and the Kelantan River Basin used statistical models and probability distributions to analyze rainfall data and reduce flood risks. These studies gave us useful information about rainfall changes and their consequences, helping policymakers and stakeholders prepare for disasters and manage water resources.

Even with all this research, there's still more to learn about rainfall in India. Our study aims to add to what we already know by analyzing extreme rainfall data using advanced models and tests. We'll look at different timeframes to understand how extreme rainfall events have changed over time. Our goal is to contribute to better strategies for dealing with extreme rainfall and making sure communities and infrastructure in India can handle it.

Objectives, study area and data set

In Chapter 2 of our study, we're diving into the annual maximum rainfall data from 36 different regions in India. Our goal is to find the best-fit probability distributions by using the Maximum Likelihood Estimator to determine their parameters. To make sure our findings are solid, we'll be putting these distributions through four goodness-of-fit tests: Kolmogorov-Smirnov, Anderson-Darling, Cramer-von Mises, and Bayesian Information Criterion. Once we've ranked and analyzed the best-fit distributions, we'll be able to determine the return periods for each region.

Now, let's talk about India's climate. It's characterized by four distinct seasons: winter, summer, the advancing monsoon, and the retreating monsoon. During winter (November to February), temperatures decrease from south to north, with frost in the north and snowfall in

the Himalayas. As we move into summer (March to May), temperatures start rising, and some regions experience the "loo" winds and pre-monsoon showers. The advancing monsoon (June to September) brings much-needed rainfall for agriculture, although its intensity varies across regions. Finally, the retreating monsoon (October to November) marks the transition to dry winter conditions and is crucial for crop harvesting.

For our analysis, we've gathered a whopping 121 years of rainfall data spanning from 1901 to 2022. We sourced this data from data.gov.in (1901-2015) and hydro.imd.gov.in (2016-2022).

Research Methodology

In Chapter 3, we dive deep into India's rainfall patterns using lots of data from the Indian Meteorological Department. We divide the data into 36 regions based on geography. Then, we analyze the data using different probability distribution models. Some of them are common, like normal(N2) and log-normal (LN2), while others are more specialized, like Pearson Type III(PIII3), log-Pearson Type III(LPIII3), Gumbel (GUM2), generalized extreme value (GEV3), Weibull(W2), generalized Pareto (GP3), and Pearson Type V distributions (PV3).

We put these models to the test with some fancy tests like Kolmogorov-Smirnov, Anderson-Darling, Cramer-von Mises, and Bayesian Information Criterion. These tests help us figure out which models are the most accurate and reliable. We also create Q-Q plots to see how well the distributions match up visually. And to look at extreme rainfall events and long-term trends, we analyse the models over different time intervals, like 5, 10, 25, 50, 100, and 200 years. Our goal is to provide useful information for policymakers, researchers, and anyone interested in managing India's water resources wisely.

Result and discussion and conclusion

In Chapter 4 of our study, we share the results and conclusions of our investigation into the rainfall patterns in 36 different regions in India. We noticed significant variations in precipitation levels, which were influenced by the geographic location of each region. To analyze these patterns, we used measures like median, standard deviation, and skewness. Through tests like K-S, AD, CVM, and BIC, we determined the best-fit probability

distributions for each region. The Maximum Likelihood Estimator and Q-Q plots helped us estimate the distribution parameters. We also calculated return periods for different time intervals.

By identifying regions with high and low mean monthly rainfall and analyzing the coefficient of skewness, we were able to understand the variability in rainfall across the regions. Each region had its own unique best-fit distribution, which led us to develop a scoring system to determine the most suitable fit. The practical outcome of our study is the calculation of return periods, which provides valuable data for flood risk modeling and prevention of heavy rainfall damage. These findings are essential for decision-making in fields like agriculture, construction, environmental management, and building architecture.

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CHAPTER-1

INTRODUCTION AND OVERVIEW

1.1 INTRODUCTION

Rainfall has a major influence on the climate of the planet. Rainfall restores ecosystems, refills water supplies like lakes and rivers, and gives both people and animals access to clean drinking water. Rainfall is essential to agriculture since farmers depend on it to cultivate their crops. Rainfall is necessary for the development and survival of trees, grass, flowers, and bushes; without it, crops would not have expanded and humans would not have access to food. Rainfall is necessary for farmers to harvest their crops. The main source of water for rivers and streams is rainfall, which aids in the production of hydropower, a clean and sustainable energy source. Rainfall is necessary to sustain wetlands, forests, aquatic plants, and other creatures, which in turn keeps ecosystems healthy. Rainfall distributes heat and moisture, which aids in controlling the temperature of the planet. It aids in lowering the ambient temperature and humidity.

Rainfall does, however, also have certain dangerous consequences. Floods caused by heavy rains can result in significant damage to houses, infrastructure, crops, and other items. Property and life are also at risk from heavy rainfall. In addition to severely eroding the soil and removing topsoil, excessive rainfall deteriorates agricultural land. Crop yields decline and biodiversity is lost as a result. Extended rainfall frequently deteriorates infrastructure, including buildings, bridges, and roads, necessitating expensive repairs. Landslides brought on by heavy rains can also cause devastation and fatalities. Rainfall can occasionally have a high pH due to contaminants in the atmosphere, causing acid rain. Acid rain degrades forests, destroys ecosystems, and lowers soil fertility. The study's objective is to determine which probability distributions suit the supplied rainfall data the best by utilizing a variety of probability distribution analysis techniques. The process of choosing a statistical distribution that best fits the available data is known as distribution fitting.

1.2 LITERATURE REVIEW

Numerous writers have examined the study of severe precipitation using various probability distributions to identify the probability distribution that best fits the data using various goodness of fit tests in various geographic locations. Here is a list of a few of their creations:

In the Northern regions of Pakistan, the study was based on a 24-period sample. They used the goodness of fit test by utilizing four Probability distributions: - normal, lognormal, log Pearson type III Gumbel max and decided that the normal distributions were found to be the best-fit Probability distribution at the Marden rainfall gauging station and Pearson type III as best fit at other gauging stations. (Amin et al. 2016). The study of the Case of the River Tiber in Rome showed the probability distribution of annual maximum peak flows on the River Tiber at the Ripetta gauge; they used data from the XV of one hundred years. They checked 2 successions: - the systematic series since 1782 and the censored series of exceptional floods from the XV century. To the optimum distribution law for high return period quantiles, an index was proposed considering both accuracy and uncertainty of estimation. (Calenda et al. 2009). The study in the United States demonstrated that record Pearson (LP3) was urged by the US water money cabinet 1957 as the base design flood repeatedly reasoning, the study told various belief systems for log Pearson type 3 distribution. The performance of the various methods and compared with the Monte Carlo simulation with the object of identifying the most robust estimator among them it was found besides the USWRC method, the method of Maximum likelihood and entropy also performed poorly. (Arora et al. 1989).

They used the probability plot correlation test statistics and L-moment diagrams to examine the complete series and wet-day series of daily precipitation records at 237 U.S stations The analysis found that the Pearson Type-III (P3) distribution fit the entire set of daily precipitation data exceptionally well, while the kappa (KAP) distribution accurately represents the observed distribution of daily rainfall on wet days. (Hanson et al. 2008). In the study analysis conducted in Malaysia, they compared eight distributions that provide accurate and reliable maximum rainfall estimates for Malaysia. The distributions that they considered were Gamma, Generalized Normal, Generalized Pareto, Generalized Extreme Value, Gumbel, Log Pearson Type III, Pearson Type III, and Wake by. They selected their study from the Annual maximum rainfall series for one-hour resolution from a network of 17 automatic gauging stations located throughout the peninsula of Malaysia. The length of rainfall records varied from 23 to 28 years. The parameters were estimated using the L-moment method. Their evaluation included the Probability Plot Correlation Coefficient test, root mean squared error, relative root mean squared error, and maximum absolute deviation. Bootstrap resampling assessed extrapolative ability. The research determined that the GEV

distribution proved to be the most suitable for representing the annual maximum rainfall in Malaysia. (Zalina et al. 2002). The study analysis conducted in Bangladesh aimed to identify the best-fitted distribution for monthly rainfall data from distantly located stations (Chittagong, Dhaka, Rajshahi, and Sylhet) spanning from 1979 to 2013. Distributions including Normal, Lognormal, Gamma, Weibull, Inverse Gaussian, and Generalized Extreme Value were assessed using the L-moment method. Goodness-of-fit test (Kolmogorov-Smirnov, Anderson-Darling, and Chi-square). Generalized Extreme Value distribution was found to be the most suitable for monthly rainfall data at three stations, while only Dhaka station showed a good fit with the Gamma distribution. (Ghosh et al. 2016).

The analysis of monthly rainfall in central west Brazil. They used eight probability distributions such as a two-parameter Weibull and, a three-parameter. Weibull, two-parameter Rayleigh, two-parameter Gamma, three-parameter Gamma, two-parameter log-normal, three-parameter lognormal and maximum Gumble for modelling the monthly rainfall data of the pluviometric station of Campo Grande, MS, Brazil, from 1975 2013. The parameters were estimated using the maximum likelihood estimation method. Six tools for assessing goodness of fit, including the chi-squared test, Kolmogorov–Smirnov test, Akaike information criterion, Bayesian information criterion, root mean square error, and coefficient of determination, were employed to determine the most appropriate probability distribution. The goodness of fit test indicated that no distribution provided the best fit to the rainfall data for all months, however, the three-parameter lognormal distribution showed a generally better fit than the other distributions, and the two-parameter log-normal distribution was the worst fit among the given distributions. (Ozonur et al. 2021). In the study analysis is from two Italian regions, i.e. Lazio and Sicily. They focused on modelling maxima daily rainfall values. Six theoretical distributions were used to evaluate their performances in fitting extreme precipitation samples. The samples were selected using two common methods, i.e. the Peak-Over-Threshold and the Annual Maxima. The Ratio Mean Square Error Method and the Kolmogorov-Smirnov test were used to assess the best-fitting distribution. Results showed that heavy-tailed distributions better described empirical data than light-tailed ones. (Moccia et al. 2021). The study analysis was conducted in southern China. The investigation in the Dongjiang River Basin in southern China focused on alterations in extreme rainfall streamflow processes. Several nonparametric techniques were

utilized for this study including the Mann-Kendall test to identify trends and the Kolmogorov Smirnov test, Levene's test, and the quantile test to detect changes in probability distribution. They observed little changes in the annual extreme precipitation series while significant changes were detected in several annual extreme flood flow and low-flow series, mainly at the stations along the main channel of Dongjiang River, which was significantly affected by the operation of several major reservoirs. The power of three non-parametric methods was assessed by Monte Carlo simulation to assess reliability. The simulation results showed that, while all three methods worked well for detecting changes in two groups of data with large sample sizes and large differences in distribution parameters, none of them was powerful enough for small data sets and small distribution parameter differences. (Wang et al.2008).

The analysis was conducted in the regions of the Brazilian Amazon, the study aimed to model the daily precipitation extreme events and estimate their return periods using generalized extreme value distribution (GEV) and the generalized Pareto distribution (GPD) in homogeneous regions of the Brazilian Amazon. The goodness of fit was assessed using quantile-quantile (Q-Q) plots and the Kolmogorov–Smirnov (KS) test, confirming the suitability of the distributions for studying extreme events. More intense precipitation extremes were expected in the southern sub-regions and coastal areas of the Brazilian Amazon. (Santos et al. 2016). The study analysis conducted in Pakistan was assessed using four distribution methods, i.e., Generalized Extreme Value (GEV), Log Pearson 3 (LP3), Gumbel Max, and Normal. Data from hydrological stations along the Swat River in Pakistan were analysed for various return periods. Three goodness-of-fit tests were applied to the fitted distributions: - Kolmogorov–Smirnov, Anderson–Darling, and Chi-squared at a 5% significance level. LP3 and GEV showed the best fit while Gumbel Max and Normal performed less favourably. LP3 was chosen for estimating flood magnitude and return periods at Khwazakhela. (Farooq et al 2018). Analysis conducted in Japan showed that the frequency analysis of annual maximum hourly rainfall of 15 locations in Japan was carried out using the Expanded Automated Meteorological Data Acquisition System (EA) weather data of 20 years from 1981 to 2008. they used eight formulas to expect the return period in years of annual maximum hourly rainfall. Five different probability distributions (PDFs) were adopted to predict the probability distribution of the occurrence of annual maximum hourly rainfall. The goodness of fit was evaluated using the Chi-square test. They showed that the Log-Pearson type 3 (LP3) distribution was the overall best fit PDF for annual

maximum hourly rainfall at most locations of Japan. (Yuan et al. 2018). The analysis concluded over the Iberian Peninsula compared two methods, peaks-over-threshold (POT) and block maxima (BM) with the generalized extreme value (GEV) distribution, for estimating future return levels (RLs) of extreme rainfall in the Iberian Peninsula (IP) using high-quality rainfall data from 1961 to 2010. They considered both all-days and rainy-days-only datasets for POT due to the prevalence of zero daily rainfall values. Rainy-days-only data were used for BM to assess trends in mean, variance, and rainy days. they showed that POT yielded higher future RL estimates as compared to GEV, attributed to significant positive trends in the number of rainy days in some regions, particularly in autumn. (Acero et al 2018).

Analyses conducted in Qatar showed that they examine the selection of best fit, probably a distribution and ring for design. They used the annual maximum rainfall data from 29 stations in Qatar, ranging from 24 to 49 years with an average of 36 years. They use 14 different distributions and three goodness-of-fit tests (Kolmogorov–Smirnov, Anderson–Darling, and Chi-squared). Using a relative scoring method, they found that the GE was the best for distribution. (Mamoon et al 2017). Studying analysis conducted on the Indus River aimed to estimate daily maximum precipitation. Using best-fit probability distribution, they observed data from 50 stations, ranging from 1961 to 2015. They used 9 distributions and parameters were estimated using the method of moments. The goodness of it test was employed and the best fit probably model was selected which leads to minimum value. (Rizwan et al. 2023). Analysis conducted in northeastern Algeria; they analysed annual maximum daily precipitation (AMDP) from 1970 to 2014 covering eight watersheds. The generalized extreme value (GEV) distribution was found as the best-fit distribution for AMDP frequency analysis. Maximum value of AMDB varied between 60 and 150 mm, with higher values in the humid zone and lower values in the arid zone. They used the Wald–Wolfowitz test for independence, the Kendall test for stationarity, and the Wilcoxon test for homogeneity. They concluded that studies identified areas suffering from extreme rainfall, urging investments in water infrastructure to handle climate change. (Benaini et al. 2023). The study analysis conducted in Turkey aimed to determine the best-fit probability distribution functions for annual rainfall. They used 10 probability distributions: gamma, generalized extreme value, Gumbel, normal, logistic, log-logistic, two-parameter log-normal, three-parameter log-normal, Pearson type III and Weibull, for modelling annual rainfall data (1975–2014) from

155-gauge stations uniformly distributed across Turkey. They evaluated the performance by various model selection techniques: Akaike information criterion (AIC), Bayesian information criterion (BIC), Anderson–Darling (AD) criterion, the Cramér–von Mises test (CVM), and the Kolmogorov–Smirnov test (KS). Using the ranking-based scoring evaluation, the gamma, log-logistic, two-parameter log-normal, normal, and Weibull distribution showed the best fit for 78% of the data. (Tosunoglu et al. 2019).

The study analysis conducted in Melbourne addresses the impact of anthropogenic language change and climate variability on extreme rainfall events. They used trend, non-stationarity analysis, and stationary and non-stationary generalized Pareto distribution (GPD) to investigate trends and non-stationarity in extreme rainfall events. Significant trends were observed in some storm durations however the analysis did not detect nonstationary but stationary models for effectively fitting the data. (Yilmaz et al. 2014). According to the American study, the writers look at the probability distribution models for the sample regions. Rainfall estimates were derived from satellite sampling using 15-minute radar rainfall data over five years. The results of the study showed that the sampling error distribution is highly dependent on the rate of rainfall; for this reason, the conditional sampling error distribution provided more information than the marginal distribution. The logistic distribution provided a good fit for the conditional sampling error at short intervals (3 or 6 hours), but the shifted-gamma, shifted-Weibull, or shifted-lognormal distributions provided a better fit for the conditional sampling error at big intervals (12 or 24 hours) than did the logistic or normal distribution. A research study in Thailand proposed combining zero inflation with an asymmetric probability distribution to effectively simulate daily rainfall data. Data from 123 rain gauges in Thailand were analysed using the following distributions: Generalized Pareto distribution, Exponential, Beta, Gamma, General extreme value, Normal, Lognormal, Weibull, and Rayleigh distributions. A density function (CDF) is created after calculating the statistical distribution using the maximum likelihood estimation (MLE) method. The Kaplan-Meier method was used to compare the CDF with the estimated CDF of the observations. Three Goodness of tests (Kolmogorov-Smirnov, Anderson-Darling, and Chi-square test) were used to evaluate the goodness of fit test. (Pakoksung et al. 2017). The goal of an analysis carried out in Kigali City was to determine how frequently and how big strong rainstorm episodes occurred there. They examined exceedances by analyzing severe climates with peaks over the threshold using a generalized parallel distribution. Intense downpour

events that often occurred in Kigali City caused serious damage to human property, infrastructural damage, human injuries, fatalities, and other negative outcomes. From 1990 until 2019, the Rwanda Meteorological Agency collected daily rainfall data from the Kigali Airport station. The returns also rise with length, suggesting that future rainfall in Kigali will become more intense and frequent. After testing the model's quality, the study concluded that a model is excellent if its shape parameter (ξ) is non-negative. These results are crucial for comprehending and addressing the incidence of intense precipitation events, supporting the formulation of policies and decision-making, and lessening their consequences (Singirankabo et al. 2022).

For nine sites in the Lake Victoria basin (LVB) in Eastern Africa, the generalized Pareto distribution (GPD) for extreme rainfall was calibrated with uncertainty using a large number of rainfall time series from global climate models in addition to observed data. It looked at the kind of GPD that might best capture the distribution's extreme quantiles and tail behaviour. To determine which approach was best, they examined many parameter estimation techniques, such as the method of moments, maximum likelihood, L-moments, and weighted linear regression in quantile plots (WLR). Uncertainty in weighting is calculated using a combination of the jackknife method and non-parametric bootstrapping. They found that the GPD test was best. Although all parameter estimation methods perform well across a wide range of tests, the WLR method appears to be more robust than other methods. The estimated uncertainty ranges from 9.25% to 59.66% of the 10-year rainfall, providing a better understanding of the risk-based planning, design, and operation of hydraulic engineering and water management associated with LVB floods. (Onyusa et al. 2015).

The analysis in Bangladesh sought to determine the best possible probability for maximum monthly rainfall using 30 years of data (1984 to 2013) from 35 locations in the country by performing multivariate analysis and classification. They estimated the parameters using three temporal and temporal evaluation and Ltime estimation techniques. The distribution with the lowest score for each of the three statistical tests was considered the best fit for each location. The distributions with the highest number of positive values are the maximum distribution, Pearson Type 3 distribution, and Log Pearson Type 3 distribution. Both Pearson type 3 and log Pearson type 3 provided the best match for 26% of sites, while the overall

model was suitable for 36% of the sites. The results of the study could improve the accuracy of flood risk and damage models. (Alam et al. 2018) developed a decision support tool (SMExRain) for the analysis of precipitation patterns based on an analysis of Canadian studies to determine the best fit to predict heavy rain. Ten models are widely used in estimating maximum annual precipitation: Beta-K, Beta-P, Generalized Extreme Value (GEV), Generalized Logistic (GLO), Generalized Normal (GNO), Generalized Pareto (GPA), Gumbel, Log Pearson Type III (LP3), Pearson Type III (P3) and Wakeby were analysed for descriptive and predictive ability. They tested the device using data from 21 rain gauges in Ontario, Canada. The indicator measures the maximum amount of precipitation every five hours, one hour per year, and every twenty-four hours. Based on various qualitative tests, GEV, GNO and P3 models have been proven to be the most effective models in explaining the annual precipitation distribution in the region. However, the GEV classification was chosen over GPA and P3 because it relies on a strong theoretical basis to reflect the distribution of different parameters and has a less common method of estimating inequality. Based on the decision support tool, GEV was determined to be the best model to describe the annual precipitation distribution in Ontario. (Nguyen et al. 2016).

Hourly precipitation observations from the Department of Meteorology and Climatology of the Aristotle University of Thessaloniki meteorological station, covering the years 1947–2003, were analysed in Thessaloniki. A novel method was applied to determine the cutoff point for classifying heavy precipitation events in Thessaloniki. The analysis of polynomial trends in the cumulative distribution functions produced from hourly rainfall data was the main focus of this methodology. To find the best-fitting probability distribution for both yearly and seasonal severe precipitation circumstances, they performed three statistical goodness-of-fit tests. determined the return times for a range of severe precipitation amounts. The hourly threshold of severe precipitation in Thessaloniki during the research period was found to be 6.5 mm/h, according to the results. It was discovered that the Generalized Pareto, Johnson SB, log-Gamma, and lognormal models matched the data on yearly and seasonal severe rainfall occurrences the best. The study also demonstrated how hourly data contributes to increased result accuracy. (Douka & associates, 2018). Research carried out in the flood-prone Kelantan River Basin revealed that the significant intermittency of northeast monsoon precipitation episodes presented difficulties for stochastic modelling

of precipitation. They created, adjusted, and verified a single-site stochastic precipitation model that incorporates both an intensity and an occurrence model for precipitation. The geographical correlation of precipitation was not taken into consideration throughout the separate simulation procedure at each location. They only took into account Markov chains with distributions up to order six and sixth. A selection of 17 rainfall stations' daily precipitation data from 1954 to 2013 was made. According to the results, modelling monthly and annual precipitation patterns was best done using second and third-order Markov chains; precipitation events were overestimated when a fifth-order Markov chain was used. For the precipitation amounts' mean, distribution, and standard deviation, the gamma, log-normal, skew-normal, mixed exponential, and generalized Pareto distributions all fared well. Nonetheless, the log-normal and exponential distributions fared better for severe precipitation than the skew-normal and extended Pareto distributions. The distribution that performed the best at approximating precipitation levels turned out to be the log-normal distribution. (Ng et al 2018).

Investigations were conducted into the performance of 8 3-parameter probability distributions for the representation of early extreme and partial duration precipitation data at stations located in the Northeastern and Southeastern regions of the United States. The extreme right tail of the yearly extreme series is best described by beta-k distribution, whereas the beta-P distribution is suitable for partial duration data. For both yearly extreme and partial duration data, it was discovered that the 2-parameter Gumble distribution significantly underestimates the probability associated with greater quantities of precipitation. The distributions did not fit the right tail any better when left-censored data was used to fit them. (Wilks et al. 1993). The authors previously presented a distribution-free goodness test for which certain large sample significant points were tabulated. The test was sensitive to differences at the tails of the distribution rather than close to the middle since they used the actual data without grouping. Birnbaum illustrated Kolmogorov's test using a numerical example. (Anderson et al. 1954). The process of choosing a model from a range of varying dimensional models involved determining its Bayes solution and assessing the leading terms inside its asymptotic expansion. Because these words were independent of the prior distribution, they constituted legitimate large-sample criteria outside of the Bayesian framework. The well-known textbook also functions as a reference tool, providing with quick, clear, and comprehensive instructions on when to use the most common non-

parametric procedures. Certain processes that had become particularly intense over time were examined such as Fisher's- Exact test, the Mantel-Haenszel Test, the Jonckheere-Terpstra Test, and the Page Test. That third device edition included updated practice questions and problems, new examples, and several updated vectors related to relevant topics. There was helpful Computer advice that may still be important in the years to come. (Conover et al. 1999).

This analysis has also been performed in India by some of the researchers. Their work has been listed below: -

Study analysis conducted in Maharashtra, aimed to determine the suitable extreme value distribution, identifying annual extreme rainfall and effectiveness of gap-filling for missing annual record/s. The suitability of different statistical distributions for representing high annual rainfall values was evaluated using the Anderson–Darling test (quantitative), quantile, and probability plots (qualitative). Gap-filling exercise resulted in minor variations (<10%) in the estimated extreme, both positive and negative across different cases. They advocated the results of EDA in the actual records. Suitable distributions for 35 districts were presented in maps and Fréchet distribution was deemed to be suitable for almost 60% of districts. The estimated extremes for all districts were listed for various return periods which are useful for the planning and construction of hydraulic structures, water resources and reservoir management, design, and planning of irrigation. (Gandhre et al 2024). The study analysis conducted in Junagadh Region (Gujarat) shows that Rain is a meager and crucial hydrological variable in arid and semi-arid regions. The area faces uncertainties in monsoon rainfall, affecting agriculture and water resources management activities. Urgent attention is needed to address water resources conservation and crop damage issues caused due to rainfall deficits or excess. They collected the daily rainfall data for a period of 38 years spanning 1984 - 2021. They fitted eight different theoretical probability distributions to the monthly and annual maximum rainfall for one to five consecutive days to select the best one for the better prediction of maximum rainfall. The goodness of fit tests including Chi-Square and Nash-Sutcliffe Efficiency were carried out by comparing the expected values with the observed values. Gumbel distribution emerged to be the best fit for the prediction of monthly and annual maximum rainfall of the Junagadh Region. (Gundalia et al 2023).

An analysis carried out in India reveals that a monthly data set covering 135 years (1871–2005) for 30 sub-divisions (sub-regions) in India was used to study precipitation patterns in India. Only three subdivisions (Coastal Karnataka, Punjab, and Haryana) saw statistically significant increases in yearly rainfall over the other half. On the other hand, out of the 15 sub-divisions exhibiting a declining tendency in annual rainfall, only one sub-division (Chhattisgarh) revealed a substantial falling trend. Contributed to more than 80% of the yearly rainfall during the monsoon months of June through September. Subdivisions with rising rainfall in June and July are equally numerous as those with decreasing rainfall. In August, there were more subdivisions with an upward tendency; in September, the reverse was seen. (Kumar et al 2010). The analysis carried out in India examined and predicted term changes in rainfall time by analyzing data from 1901 to 2015 at a meteorological divisional level across the country. They employed the Pettitt test to detect the abrupt change point in the time frame, the Mann-Kendall (MK) test, and Sen's Innovative trend analysis to analyze the rainfall trend. They forecasted the 15-year rainfall across India using Artificial Neural Network-Multilayer Perceptron (ANN-MLP) and mapped the trend in rainfall for the whole country by using the geo-statistical technique like Kriging in ArcGIS environment. Their results showed that except for 7 divisions rest, all faced a negative trend of rainfall in annual and seasonal scales. Out of 17 divisions, 11 divisions recorded a noteworthy declining trend in the monsoon season at a 0.05% significance level, while insignificant negative trends of rainfall were detected for the winter and pre-monsoon seasons. Furthermore, a significant negative trend (-8.5) was recorded for overall annual rainfall. Increasing rainfall trends were observed from 1901 to 1950 while a significant decline in rainfall was detected after 1951. Rainfall forecasts for the next 15 years for all the meteorological divisions also exhibited a significant decline in rainfall. They derived their results from ECMWF ERA5 reanalysis data that exhibited increasing/decreasing precipitation convective rate, elevated low cloud cover and inadequate vertically integrated moisture divergence might have influenced the change of rainfall in India. (Praveen et al 2020).

Examining the study carried out in Roorkee reveals that rainfall is a significant hydrological variable in arid regions, and its importance is growing as a result of urbanization, economic expansion, and population growth. Additionally, they contend that probability analysis is necessary to resolve a variety of water management concerns and guard against agricultural

damage brought on by little or excessive rainfall. Their research sought to examine how frequently certain rainfall data occurred. The maximum annual rainfall was predicted using a variety of probability distributions, including the Gumbel, Log Normal, Log Pearson Type III, and Van Te Chow tables. They obtained annual rainfall data from the National Institute of Hydrology (NH), Roorkee, India, spanning 20 years (1991–2010). By choosing the optimal distribution for each scenario, the research tried to match these distributions to the maximum yearly rainfall for daily, two consecutive days, and three consecutive days. To compare the predicted and observed values, the goodness of fit metrics including chi-square, percentage absolute deviation, and integral square error was used. For the creation of the yearly maximum rainfall values in Roorkee for one, two, and three consecutive days, the Gamble distribution proved to be the best fit. Ven Te Chow distribution was the second-best fit for the same duration. (Kaur et al. 2021). Studying the analysis conducted in Roorkee informs us that rainfall is an important hydrological variable in dryland areas, which is increasingly becoming crucial due to population growth, economic development, and urbanization. They also suggest that probability analysis is needed to address a number of water management issues and to prevent crop damage due to deficits or excess rainfall. Their study aimed to analyse the frequency of different rainfall data. Various probability distributions namely, Gumbel, Log Normal, Log Pearson Type III, and Van Te Chow were used for the prediction of the maximum annual rainfall. They collected annual rainfall data for 20 years (1991-2010) from the National Institute of Hydrology (NH), Roorkee, India. The research attempted to fit these distributions to the maximum annual rainfall for daily, two consecutive days, and three consecutive days, selecting the best distribution for each scenario. Goodness of fit such as chi-square, percentage absolute deviation, and integral square error was employed to compare the expected values with the observed values. Gamble distribution emerged to be the best fit for the production of annual maximum rainfall values of Roorkee for one, two, and three consecutive days and the second-best fit was Ven te-14 Chow distribution for one, two, and three consecutive days. (Kaur et al. 2021). Given its varied topography and intense rainfall that frequently causes quick flow and puts major constructions and other natural resources in danger, the study analysis done in Uttarakhand, India, indicates that knowing rainfall patterns is a crucial issue for the state. The objective of their research was to ascertain which distribution best suited the yearly rainfall data set covering 13 districts in Uttarakhand from

1991 to 2002. Applying several distributions to the data included Chi-squared, Chi-squared (2P), exponential, exponential (2P), gamma, gamma (3P), gen. extreme value (GEV), log-Pearson 3, Weibull, and Weibull (3P). The best distributions were compared using goodness-of-fit tests such as Chi-squared, Anderson–Darling, and Kolmogorov–Smirnov. Findings showed that the Weibull distribution, which covered 46% of the district overall, performed best; log-Pearson and Chi-squared (2P) fared second best. (Kumar et al. 2017).

According to the examination of Chhattisgarh, climatic variability and its behaviour presented a complicated phenomenon full of unknowns. Finding characteristics (predictors) that are either directly or indirectly connected to predictands is important in the study of climate change, especially as it relates to hydrological elements. In this investigation, the daily rainfall in the Piperiya watershed of the state of Chhattisgarh was downscaled using the statistical downscaling model (SDSM), which is based on the multiple linear regression approach. The model's input data were daily precipitation records from 1961 to 2001 from Chhattisgarh's Piper watershed. Rainfall data from 1961 to 1990 and 1991 to 2001, along with large-scale predictors from reanalysis data from the National Centre for Environmental Prediction (NCEP), were used to calibrate and verify the model. On the basis of the anticipated daily rainfall for each period, monthly rainfall was then estimated. Sagar Island analysis revealed that it is one of the most vulnerable deltas to severe rainfall-induced climate risks. Their study sought to offer insightful information on the likelihood of monthly, seasonal, and yearly maximum daily rainfall events (MRD). Six probability distribution models were investigated through the analysis of daily rainfall data collected over a period of 2022–2010. For yearly, seasonal, and monthly time series, the optimal distribution models were chosen by maximizing rank and reducing test statistic value. Three tests of statistical goodness of fitness were employed, including the Chi-Square test (X^2), the Anderson Darling test (A^2), and the Kolmogorov–Smirnov test (K-S). From the three goodness of fit tests, the fourth probability distribution with the greatest score was identified. The results show that the normal distribution is best for all years for post-monsoon, and summer MDR, while poor season Weibull and person 5 are best for pre-monsoon, monsoon and winter. Estimated annual MDRs for return periods of 2, 5, 10, 20 and 25 are

50, 69, 86, 106 and 114 mm, respectively, and the probability of receiving an MDR each year is > 50 , > 100 , > 150 , > 200 and > 250 mm is approximately 99%, 85%, 40%, 12% and

0.3% above standards. In summer, summer and winter, the incidence of MDR > 100 mm is high (78% to 85%), while the incidence of MDR > 150 mm is moderate (37% to 46%). (Mandal et al. 2015).

The study analysis conducted at the Kelantan River Basin experiences significant monsoon seasons particularly Northeast and Southwest monsoons that lead to flood and heavy downpour events. The aim of their study was to analyse the Annual Maximum Series (AMS) and Partial Duration Series (PDS) by fitting different probability distributions. Various distributions such as: - Generalized Extreme Value (GEV), Generalized Pareto (GP), Log Pearson Type 3 (LP3), Log Normal (LN) and Log Normal 3 (LN3) were used. The performances of these probability distributions were evaluated using different goodness-of-fit tests such as chi-square (χ^2), Kolmogorov-Smirnov (KS), and Anderson-Darling (AD) tests. The GEV distribution was found to be the best fit for the AMS, while the GP distribution was identified as the best fit for the PDS. These findings offer valuable insights for flood mitigation efforts in the region. (Ng et al. 2021).

In order to determine the origin of the 36 meteorological regions of the Indian Meteorological Department (IMD), this study will analyze the extreme rainfall data using the best-fit probability distribution with recursive intervals of 5, 10, 25, 50, 100, and 200 years, as well as the best goodness of fit tests.

CHAPTER - 2

OBJECTIVES, STUDY AREA AND DATA SET.

2.1 OBJECTIVES

- We will be using the data collected from annual maximum rainfall data for 36 meteorological regions.
- From the literature review, we will be identifying the best fit probability distributions.
- The Maximum Likelihood Estimator will be used to identify the parameters of all the probability distributions selected.
- We will be using four Goodness-of-fit tests i.e. K-S, A-D, CVM and BIC to identify the best-fit probability distributions for each region.
- We will be using the Ranking method to identify the best fit probability distribution.
- We will be analyzing the best-fit probability distributions identified in the literature review.
- On the basis of best fit probability distribution, we will then determine the return periods for 5, 10, 25, 50, 100 and 200 years across all the regions.

2.2 STUDY AREA AND DATA SET

India experiences a range of weather conditions due to its vast size, varied topography and monsoonal climate. India experiences four distinct seasons - the cold weather season, the hot weather season, and the advancing and retreating monsoons, with some variations across different regions.

- **The Cold Weather Season (Winter):** - The winter season in northern India begins from mid-November and lasts until February. December and January are the coldest months during this period. As we move from South to North, temperature decreases gradually. Days are warm, but nights are cold. Frost is common in the northern regions, and at the higher area, such as Himalayas, often experiences snowfall. During this season, the weather has clear skies, low temperatures, low humidity and variable winds.
- **The Hot Weather Season (Summer):** - The hot weather season in India begins from March to May. During these months, the temperature rises and air pressure decreases in the northern part of the country. This weather also experiences the occurrence of 'loo' winds, strong, gusty, hot, dry winds blowing during the day over the north and

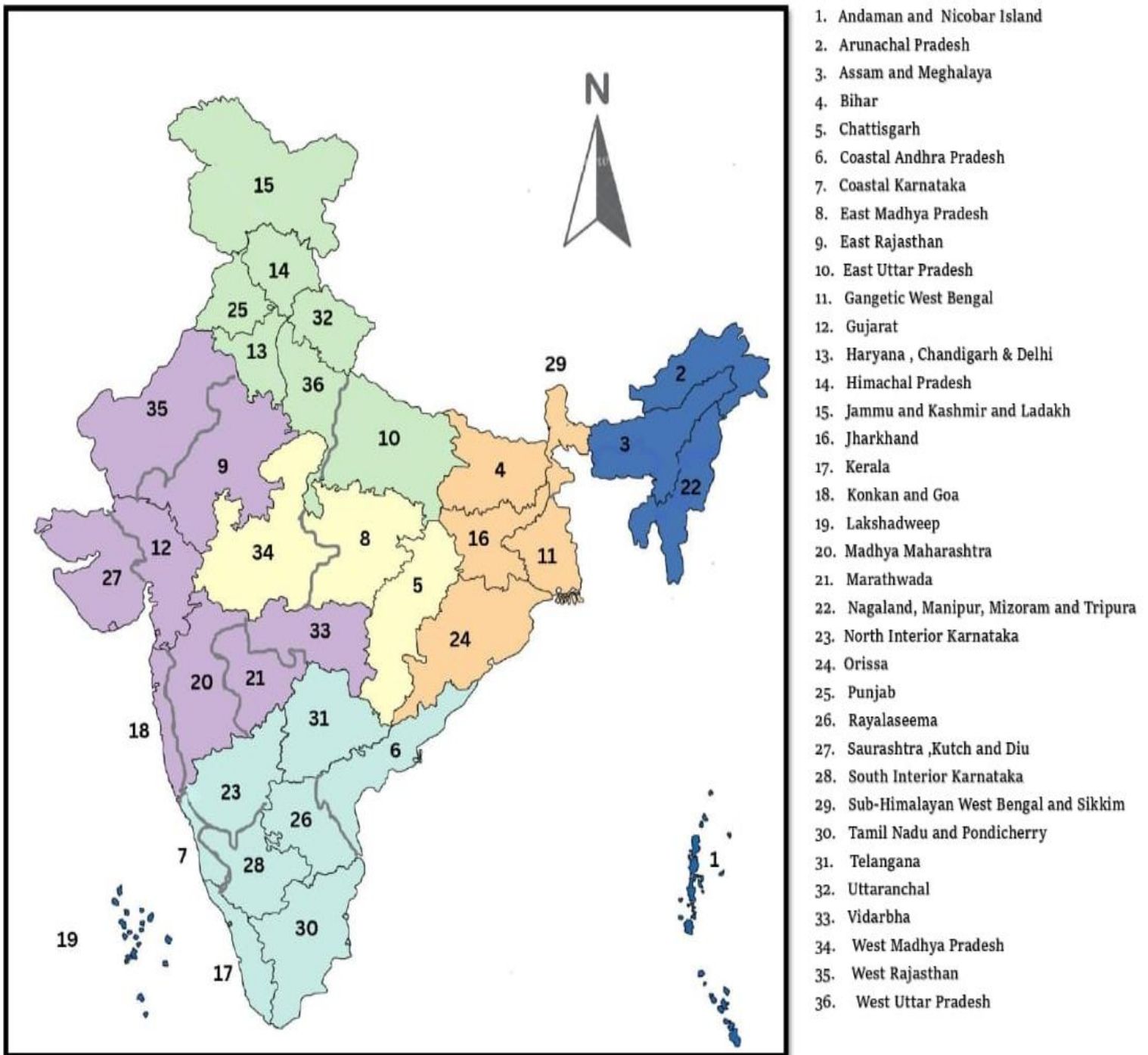
northwestern India. During this period certain regions often experiences the beginning of drought conditions. Towards the end of summer season pre-monsoon showers are also visible, especially in cereal and Karnataka. These are known as mangoes showers, which helping in ripening of mangoes.

- **The Advancing Monsoon (Rainy Seasons):** - The rainy season in India spans from June to September, which is characterized by the arrival of Southwest monsoon winds, which brings rainfall across the country. The onset of the monsoon widespread showers, replenishing water resources and supporting agricultural activities. Rainfall is very crucial for individual economy, particularly for the agricultural sector as it provides moisture to the crops for their survival and growth. However, rainfall intensity can vary from region to region. Heavy rainfall often leads to floods in some areas while other regions might experience droughts due to water scarcity. Despite these Challenges, rainy season is welcomed in India as it provides relief from summer heat.
- **Retreating/Post Monsoons (The Transition Season):** - The retreating monsoon often known as post monsoon or transition season occurs between October – November. During this season, the south-west monsoon winds become weaken and starts withdrawing gradually. The withdrawal of monsoon from the Northern Plains starts in beginning of October. The months of October-November mark a transition from the hot rainy season to dry winter conditions. Days experiences high temperatures while nights are cool and pleasant. The retreating monsoon season is characterized by changing weather patterns, with a gradual shift towards cooler temperatures and drier conditions. This season also plays a crucial role in India's agricultural cycle, as it marks the end of the rainy season and the beginning of the harvesting period for many crops. Farmers use this time to prepare their fields for the upcoming winter crops and to complete the harvesting of crops planted during the monsoon season.

The data was collected for 36 different regions across all over the country. This analysis uses 121 years of data from 1901-2022. The data for the years 1901-2015 was taken from data.gov.in and the data for the years 2016 – 2022 was taken from hydro.imd.gov.in. The

data was recorded as the annual maximum rainfall at each location. The following figure, Figure 1, displays the 36 meteorological regions: -

Figure: - 1
36 Meteorological Regions in India



CHAPTER-3

RESEARCH

METHODOLOGY

3.1 METHODOLOGY

In this study, we have used 9 best-fit probability distribution based on Goodness-of-fit tests i.e. K-S, AD, CVM and BIC on 36 regions for extreme rainfall analysis and have been presented below along with their probability density functions, cumulative distribution functions and their parameters: -

3.1.1. Commonly used Probability Distributions

This analysis adopts commonly used frequency distributions used in our study above. On the basis of the literature review, we have analyzed and selected 9 best distribution models that fits the extreme rainfall data in our case. The parameters of the distributions have been estimated using method of **Maximum Likelihood Estimator (MLE)**. The selected distribution models that are commonly used in extreme rainfall analyses are presented below:

-

1) Normal Distribution (N2)

The Gaussian or N distribution is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{1}{2\sigma^2}(x-\mu)^2\right]}$$
$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\exp \left[-\frac{1}{2\sigma^2} (x - \mu)^2 \right] \right) dx$$

This distribution has two parameters: - μ is the mean of the distribution and σ^2 is the variance

of the distribution with a range of $-\infty < \mu < \infty$, $0 < \sigma < \infty$

2) Lognormal Distribution (LN2)

This distribution is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{1}{x\sigma_Y\sqrt{2\pi}} \exp\left[-\frac{1}{2\sigma_Y^2}(\ln(x) - \mu_Y)^2\right]$$
$$F(x) = \frac{1}{\sigma_Y\sqrt{2\pi}} \int_0^x \left(\frac{1}{x} \exp\left[-\frac{1}{2\sigma_Y^2}(\ln(x) - \mu_Y)^2\right]\right) dx$$

This distribution has two parameters: - μ_Y is the mean and σ_Y is the standard deviation

With a range of $\mu \in (-\infty, \infty)$, $\sigma \in (0, \infty)$

3) Pearson Type 3 Distribution (PIII3)

This is one of the most commonly used distribution in hydrology and is it 2-parameter gamma distribution, its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{1}{|\alpha|\Gamma\beta} \left[\left(\frac{x-\xi}{\alpha}\right)^{\beta-1}\right] \exp\left[-\left(\frac{x-\xi}{\alpha}\right)\right]$$
$$F(x) = \frac{1}{|\alpha|\Gamma\beta} \int_{\xi}^x \left[\left(\frac{x-\xi}{\alpha}\right)^{\beta-1}\right] * \exp\left[-\left(\frac{x-\xi}{\alpha}\right)\right] dx$$

This distribution has three parameters: - α is the scale parameter, β is the shape parameter and ξ is the location parameter with a range of $\alpha > 0$, $\beta > 0$, $0 < \xi < x$

4) Log-Pearson Type 3 Distribution (LPIII3)

This is another form of gamma family distribution. Its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{1}{|\alpha|x\Gamma\beta} \left[\left(\frac{\ln x - \xi}{\alpha} \right)^{\beta-1} \right] \exp \left[- \left(\frac{\ln(x) - \xi}{\alpha} \right) \right]$$

$$F(x) = \frac{1}{|\alpha|\Gamma\beta} \int_0^x \frac{1}{x} \left(\frac{\ln(x) - \xi}{\alpha} \right)^{\beta-1} \exp \left[- \left(\frac{\ln(x) - \xi}{\alpha} \right) \right] dx$$

This distribution has three parameters: - α is the scale parameter, β is the shape parameter and ξ is the location parameter with a range of $\alpha > 0$, $\beta \neq 0$, $0 < \xi < x$

5) Pearson Type 5 Distribution (PV3)

The Pearson Type 5 distribution is also often called the **Inverse Gamma distribution** is often applied in extreme rainfall analysis; its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{e^{-\frac{\beta}{x-\gamma}}}{\beta\Gamma\alpha \left(\frac{x-\gamma}{\beta} \right)^{\alpha+1}}$$

$$F(x) = \int_{\gamma}^{\infty} \left(\frac{e^{-\frac{\beta}{x-\gamma}}}{\beta\Gamma\alpha \left(\frac{x-\gamma}{\beta} \right)^{\alpha+1}} \right) dx$$

This distribution has three parameters: - α is the shape parameter, β is the scale parameter and γ is the location parameter with a range of $\alpha > 0$, $\beta > 0$, $\gamma \neq 0$

6) Gumbel Distribution (GUM2)

The extreme value type I distribution also called as Gumbel distribution is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$f(x) = \left(\frac{1}{\alpha} \right) \exp \left[- \left(\frac{x-\beta}{\alpha} \right) - \exp \left(- \left(\frac{x-\beta}{\alpha} \right) \right) \right]$$

$$F(x) = \exp \left[- \exp \left(- \frac{x-\beta}{\alpha} \right) \right]$$

This distribution has two parameters: - α is the scale parameter and β is the location parameter

With a range of $\beta \in (-\infty, \infty)$, $\alpha > 0$

7) Generalized extreme value distribution (GEV3)

A well-known three parameter distribution for maxima is GEV and is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$f(x) = \frac{1}{\beta} \left(1 + \gamma \left(\frac{x - \alpha}{\beta} \right) \right)^{-\left(1 + \frac{1}{\gamma}\right)} * \exp \left(- \left(1 + \gamma \left(\frac{x - \alpha}{\beta} \right) \right)^{\frac{1}{\gamma}} \right)$$

$$F(x) = \exp \left\{ - \left[1 + \gamma \left(\frac{x - \alpha}{\beta} \right) \right]^{\frac{1}{\gamma}} \right\}, \text{ if } \gamma \neq 0$$

$$= \exp \left\{ - \exp \left[- \left(\frac{x - \alpha}{\beta} \right) \right] \right\}, \gamma = 0$$

Three parameters of this distribution are: α is the location parameter, β is the scale parameter, and γ is the shape parameter. The range of these parameters is $-\infty < \alpha < \infty$, $\beta > 0$, $-\infty < \gamma < \infty$. The parameters α and β just rescale(shift and stretch) the distribution. They are analogous to (but do not usually correspond to) the mean and the standard deviation. The parameter γ determines the overall shape of the distribution (analogous to the skewness) and its sign (positive, negative or zero) results in three different shaped distributions.

8) Weibull Distribution (W2)

The extreme value type 3 or Weibull is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$f(x) = \left(\frac{k}{\alpha} \right) \left(\frac{x}{\alpha} \right)^{k-1} \exp \left\{ \exp \left(- \left(\frac{x}{\alpha} \right)^k \right) \right\}$$

$$F(x) = 1 - \exp \left\{ \exp \left(- \left(\frac{x}{\alpha} \right)^k \right) \right\}$$

This distribution has two parameters: - α is the shape parameter and k is the scale parameter with a range of $\alpha > 0, k > 0$.

9) Generalized Pareto distribution (GP3)

A well-known three parameter distribution for threshold exceedances is called GPD which is often applied in extreme rainfall analysis, its probability density function and cumulative distribution function is given as: -

$$\begin{aligned} f(x) &= \frac{1}{\sigma} \left(1 - k \left(\frac{x - \xi}{\sigma} \right) \right)^{\frac{1}{k} - 1}, \quad k \neq 0 \\ &= \frac{1}{\sigma} \exp \left(- \frac{x - \xi}{\sigma} \right), \quad k = 0 \\ F(x) &= 1 - \left(1 - k \left(\frac{x - \xi}{\sigma} \right)^{\frac{1}{k}} \right), \quad k \neq 0 \\ &= 1 - \exp \left(- \frac{x - \xi}{\sigma} \right), \quad k = 0 \end{aligned}$$

This distribution has two parameters: - α is the shape parameter and β is the scale parameter

With a range of $k \in (-\infty, \infty), \xi \in (-\infty, \infty), \sigma > 0$

3.1.1.1. Estimation of Parameter

The parameters of the distribution have been estimated using the Maximum Likelihood Estimator (MLE) method. Given some observed data, maximum likelihood estimator is a technique for estimating the parameters of the assumed probability distribution. This is accomplished by optimizing the likelihood function to make the observed data the most likely under the presumptive statistical model. The maximum likelihood estimate is the location in the parameter space where the likelihood function is maximized. Maximum likelihood is a

popular technique for statistical inference because of its flexible and intuitive reasoning. The MLE is obtained as: -

$$L = f(x_1, x_2, \dots, x_n | \theta) = f(x_1 | \theta) * f(x_2 | \theta) * \dots * f(x_n | \theta)$$

The average log-likelihood function is easier to work with: -

$$\hat{l} = \frac{1}{n} \log L = \frac{1}{n} \sum_{i=1}^n \log f(x_i | \theta)$$

3.1.2. Goodness-of-Fit Test

The statistical model's power comes from its capacity to illustrate how well it fits a particular collection of data. A goodness of fit usually provides an overview of the discrepancy between the actual and predicted values for the given model. These measurements can be applied to statistical hypotheses to check if the residuals are normal, whether two samples are taken from the same distribution, or whether the result frequencies correspond to a given distribution. The lack-of-fit sum of squares may be one of the components in the variance analysis into which the variance is partitioned. Three approaches are employed to verify the normalcy: formal normality tests, numerical methods (skewness and kurtosis indices), and graphic methods (histogram, boxplots, and Quantile-Quantile plots, or Q-Q plots). Numerous methods are available to verify normality in statistical testing processes. Among them are tests known as Empirical Distribution Functions (EDF), which offer a measurement of the difference between theoretical and actual distributions. The **Kolmogorov-Smirnov (K-S)**, **Anderson-Darling (A-D)**, and **Cramer-von-Mises and Bayesian information criterion (BIC) tests** are the most well-known EDF tests.

3.1.2.1 Kolmogorov-Smirnov (K-S) test

The non-parametric Kolmogorov-Smirnov test, or "**KS test**," is used to compare probability distributions. It doesn't make any assumptions on the underlying distributions of the data

because it is a non-parametric test. The distance between the empirical distribution functions of two samples, or between the empirical distribution functions of a sample and the cumulative distribution function of a reference distribution, is measured using the Kolmogorov-Smirnov statistics. The analysis is frequently used to evaluate if the distribution of data is normally distributed. It is also used in the analysis of variance to confirm the normalcy assumption. A goodness of fit test may be created by modifying the Kolmogorov-Smirnov test. Studies have shown that the Shapiro-Wilk and Anderson-Darling (AD) tests are more effective in determining normalcy than the test, even when it is adjusted. The data is rearranged in increasing order for a sample of size n , $X_1 < X_2 < \dots < X_n$ and the K-S test is assessed for each ordered value:-

$$S_n(x) = 0, \text{ if } X < X_1$$

$$= \frac{k}{n}, \text{ if } X_k \leq X < X_{k+1}$$

$$= 1, \text{ if } X > X_n$$

$$D_n = \max |F_x(x) - S_n(x)|$$

$$P(D_n \leq D_n^\alpha) = 1 - \alpha$$

Where D_n^α is the critical value, α is the significance level and k are the rank order of the data set. Ozonur (et al. 2021) and Moccia (et al. 2021)

3.1.2.2 Anderson–Darling (A-D) test

The Anderson–Darling test, known for its high efficacy compared to other Empirical Distribution Function (EDF) tests, serves as a statistical method to determine if a specific dataset originates from a particular probability distribution. In its fundamental state, this test operates under the assumption that no parameters require estimation within the distribution under examination, leading to the test and its associated critical values being independent of distribution. This test is commonly applied within scenarios where a group of distributions is being scrutinized, necessitating the estimation of parameters within that group. When this occurs, adjustments either to the test-statistics or critical values must consider this factor.

This test stands out as a potent statistical instrument for identifying deviations from normality. It is a goodness-of-fit test which allows to control the hypothesis that the distribution of a random variable observed in a sample follows a certain theoretical distribution. The test is grounded on the idea that if a data set is typically distributed, also the maximum difference between the accretive distribution function of the data and the normal distribution should be minimized. The AD test statistics, the quadratic class of EDF test statistics is expressed as A^2 as follows: -

$$A^2 = - \sum_{i=1}^n \{ (2i-1) \ln F_X(x_i) + \ln[1 - F_X(x_{n+1-i})] \} / n - n$$

Where $F_X(x)$ is the cuff of the proposed distribution at x_i , for $i = 1, 2, 3, \dots, n$. The data found (i.e. observed) must be arranged in increasing order as $x_1 < x_2 < \dots < x_n$. This test give more weights to the tails. Farooq (et al. 2018) and Mamoon (et al. 2017)

3.1.2.3 Cramer-von-Mises (CVM) test

The **Cramer-von-Mises** is a criterion used for judging the goodness of fit of a cumulative distribution function F compared to a given empirical distribution function F_n^* , or for comparing two empirical distributions. It is also used as a part of other algorithms such as minimum distance estimation Tosunoglu (et al. 2019). It is defined as: -

$$\omega^2 = \int_{-\infty}^{\infty} [F_n(x) - F^*(x)]^2 dF^*(x)$$

Let $x_1, x_2, \dots, x_n$ be the observed values, in increasing order. Then the test statistic is: -

$$T = n\omega^2 = \frac{1}{12n} + \sum_{i=1}^n \left[\frac{(2i-1)}{2n} - F(x_i) \right]^2$$

3.1.2.4 Bayesian Information criterion (BIC)

The Bayesian Information criteria (BIC), also known as the Schwarz Information Criterion (also SIC, SBC, and SBIC), is a criterion for choosing a model from a restricted collection

of models. It is closely linked to the Akaike Information Criteria (AIC) and partially based on the liability function. A statistical criterion that helps in model selection and thesis testing is the Bayesian information criterion. This method seeks to balance model fit and complexity in order to identify the fashionable, economical, and informative model. Models with a lower BIC are typically chosen. Advanced parameter models are penalized by the BIC. Ozonur (et al. 2021) and Tosunoglu (et al. 2019). Formally, the BIC is described as: -

$$BIC = k \ln (n) - 2 \ln(\hat{L})$$

Where

- $\hat{L}$ = the maximized value of the likelihood function of the model M, i.e. $\hat{L} = p(x|\hat{\theta}, M)$
Where $\hat{\theta}$ are the parameter values that maximize the likelihood function and x is the observed data;
- n = the number of data points in x, the number of observations or equivalently, the sample size;
- k = the number of parameters estimated by the model

3.1.2.5 Quantile-Quantile(Q-Q) Plots

One of the simplest and most effective methods for choosing the best fit model is a graphical test. The Q-Q plot of the estimated and observed value is shown here. A variety of plotting position formula are utilized for the particular non-exceedance probability π to determine the charting position of the distributions. The majority of the plotting position formulas in use today use the following format: -

$$P_{i:n} = \frac{n - a}{N + 1 - 2a}$$

Where $n = 1, 2, 3, \dots, N$ is the rank of the observed value X ($X(i)$ is the ascending order). N is the total number of observed values and a is a constant. For N2, LN2, PIII3, LPIII3 and PV3 distributions, Blom's plotting position ($a=3/8$) which provides unbiased quantiles, is employed here; for GUM2, W2, and GP3 distributions, Gringorten's plotting position ($a =$

0.44) is used; and GEV, Cunnane's plotting position ($a = 0.4$) is utilized (Stendinger et al. 1993). The observed data, $X(i)$ and the estimated values, $x(F)$ are displayed to create the Q-Q plot. F is defined by the $p_{i:n}$ for the certain probability distribution.

3.1.3 Return Period

The determination of the recurrence interval, often known as the return period, is one of the main goals of frequency analysis. If variable (x) is equal to or greater than an event size x_T and it happens once every T year, then $P(X \geq x)$ is the chance that the variable will occur in a given day

$$P(x \geq x_T) = \frac{1}{T}$$

$$T = \frac{1}{1 - P(x \leq x_T)}$$

It is not possible to estimate the precipitation levels associated with the 50- or 100-year average return periods directly from a dataset; instead, they must be retrieved from the fitted distribution's 98th and 99th percentiles, respectively. (Wilks et al. 1993).

CHAPTER-4

RESULTS

AND DISCUSSION

4.1 RESULTS

The primary objective of this study is to determine which probability distribution when fitted to each station, produces the highest monthly rainfall during the return periods 5, 10, 25, 50, 100 and 200 years. These approximations offer valuable direction for formulating policies and making decisions. Being aware that the return period of catastrophic and extreme occurrences can be used to gauge the degree of danger that extreme events – like floods, heavy rain, etc. will cause harm.

The parameters of the distribution have been estimated using the Maximum Likelihood Estimator (MLE). Using our nine probability distributions that we have analyzed in the literature study, we calculated the values of the parameters for each of the 36 regions that were available to us, as indicated in the Table 1.

Table 1: Distributions corresponding to all regions and their parameter values

S.No.	Region Name	Distribution Name	Values of parameters
1	Andaman and Nicobar Islands	N2	$\mu = 604.4552, \sigma = 130.8927$
		LN2	$\mu = 6.3821204, \sigma = 0.2084812$
		PIII3	$\beta=2.943522, \xi=372.949304, \alpha=78.640311$
		LPIII3	$\beta=8.407349, \alpha= 13.650221, \xi=5.766208$
		PV3	$\alpha=10.3340, \gamma=202.1843, \beta=3762.0802$
		GUM2	$\beta = 544.1362, \alpha = 102.0976$
		GEV3	$\alpha = 541.96289842, \beta = 100.44032154, \gamma = -0.03992951$
		W2	$\alpha = 4.707291, k = 658.170322$
		GP3	$\xi=391.9995506, \sigma=334.7577098, k=-0.5623618$
2	Arunachal Pradesh	N2	$\mu = 802.3817, \sigma = 361.1572$
		LN2	$\mu = 6.5952103, \sigma = 0.4253643$
		PIII3	$\beta=1.96812, \xi=287.37574, \alpha=261.67390$
		LPIII3	$\beta=15.265868, \alpha= 9.077539, \xi=4.913492$
		PV3	$\alpha= 5.278224, \gamma= 40.278593, \beta=3290.612949$
		GUM2	$\beta= 642.2322, \alpha = 260.6344$
		GEV3	$\alpha = 615.1678045, \beta = 235.5351051, \gamma= -0.2022285$

		W2	$\alpha = 2.352706, k = 908.184462$
		GP3	$\xi=302, \sigma=621.26161, k= -0.25242$
3	Assam and Meghalaya	N2	$\mu = 802.3817, \sigma = 361.1572$
		LN2	$\mu = 6.3427902, \sigma = 0.1796319$
		PIII3	$\beta=11.23065, \xi=224.68746, \alpha=31.43587$
		LPIII3	$\beta=309.259330, \alpha= 97.980266, \xi= 3.186452$
		PV3	$\alpha=10.77364, \gamma=212.93702, \beta=3594.66983$
		GUM2	$\beta = 528.2568, \alpha = 90.6444$
		GEV3	$\alpha = 532.40978402, \beta = 91.80263337, \gamma= 0.08504918$
		W2	$\alpha = 5.262063, k = 622.817992$
		GP3	$\xi=355.5, \sigma=303.74217, k= -0.46068$
4	Bihar	N2	$\mu = 369.55902, \sigma = 78.34585$
		LN2	$\mu = 5.8886664, \sigma = 0.2214601$
		PIII3	$\beta=2.943522, \xi=372.949304, \alpha=78.640311$
		LPIII3	$\beta=310.041957, \alpha=78.473031, \xi= 1.937654$
		PV3	$\alpha=13.95445, \gamma=46.62547, \beta=4210.67858$
		GUM2	$\beta = 330.74206, \alpha = 74.67814$
		GEV3	$\alpha = 340.6492266, \beta = 76.8287227, \gamma= 0.2474347$
		W2	$\alpha = 5.157906, k = 401.184048$

		GP3	$\xi=190, \sigma= 571.513, k= -1.465$
5	Chattisgarh	N2	$\mu = 440.71967, \sigma = 75.02809$
		LN2	$\mu = 6.0733522, \sigma = 0.1759269$
		PIII3	$\beta=30.63845, \xi=17.65210,$ $\alpha=13.80845$
		LPIII3	$\beta=388.367296, \alpha=110.844645,$ $\xi=2.569559$
		PV3	$\alpha=13.80675, \gamma=120.62018,$ $\beta=4135.22567$
		GUM2	$\beta= 403.40083, \alpha =73.58331$
		GEV3	$\alpha = 413.7846538, \beta= 74.5490695,$ $\gamma=0.2679801$
		W2	$\alpha = 6.382704, k= 472.382292$
		GP3	$\xi=235, \sigma=732.5815, k= -1.8518$
6	Coastal Andhra Pradesh	N2	$\mu = 440.71967, \sigma = 75.02809$
		LN2	$\mu = 5.5300243, \sigma = 0.2409161$
		PIII3	$\beta=3.700679, \xi=132.997046,$ $\alpha=34.271437$
		LPIII3	$\beta=20.396696, \alpha=18.713499,$ $\xi=4.440072$
		PV3	$\alpha=15.07832, \gamma=21.64696$ $, \beta=3351.15733$
		GUM2	$\beta= 230.08910, \alpha = 50.70933$
		GEV3	$\alpha = 229.19199150, \beta=$ $50.12496495, \gamma= -0.03267535$
		W2	$\alpha = 3.811495, k= 285.631478$

		GP3	$\xi=149.5994173$, $\sigma=169.4857901$, $k= -0.4336965$
7	Coastal Karnataka	N2	$\mu = 1180.8525$, $\sigma = 243.5707$
		LN2	$\mu = 7.0534061$, $\sigma = 0.2021338$
		PIII3	$\beta=8.14168$, $\xi=486.04058$, $\alpha=85.34013$
		LPIII3	$\beta=611.541354$, $\alpha=122.421041$, $\xi=2.058017$
		PV3	$\alpha=16.79467$, $\gamma=195.64227$, $\beta=15584.87592$
		GUM2	$\beta= 1066.876$, $\alpha=204.264$
		GEV3	$\alpha = 1.076202e+03$, $\beta=$ $2.084541e+02$, $\gamma= 8.472323e-02$
		W2	$\alpha = 4.930708$, $k= 1281.483174$
		GP3	$\xi=699.4991671$, $\sigma=848.8349795$, $k= -0.6988819$
8	East Madhya Pradesh	N2	$\mu = 432.8246$, $\sigma = 96.9358$
		LN2	$\mu = 6.0440571$, $\sigma = 0.2334784$
		PIII3	$\beta=17.70269$, $\xi=16.05644$, $\alpha=23.54304$
		LPIII3	$\beta=402.886809$, $\alpha= 85.073421$, $\xi=1.308219$
		PV3	$\alpha=13.19742$, $\gamma=46.02094$, $\beta=$ 4750.86414
		GUM2	$\beta=385.17452$, $\alpha =91.27228$
		GEV3	$\alpha = 395.8164382$, $\beta= 93.5099816$, $\gamma= 0.2177612$
		W2	$\alpha = 4.826259$, $k= 471.480088$

		GP3	$\xi=205.8995538$, $\sigma=318.5249161$, $k= -0.6205199$
9	East Rajasthan	N2	$\mu = 278.71721$, $\sigma = 72.46235$
		LN2	$\mu =5.5921932$, $\sigma = 0.2867893$
		PIII3	$\beta=10.64666$, $\xi=25.53486$, $\alpha=23.78662$
		LPIII3	$\beta=198.10786$, $\alpha=47.90447$, $\xi=1.45624$
		PV3	$\alpha=9.58604$, $\gamma=14.33784$ $\beta=2302.48749$
		GUM2	$\beta= 242.3038$, $\alpha =71.4306$
		GEV3	$\alpha = 254.0071331$, $\beta= 73.3119317$, $\gamma= 0.3065628$
		W2	$\alpha = 4.29401$, $k= 306.16518$
		GP3	$\xi=108.7998531$, $\sigma=250.4033155$ $k= -0.3969434$
10	East Uttar Pradesh	N2	$\mu = 336.59754$, $\sigma = 78.07778$
		LN2	$\mu = 5.7919891$, $\sigma = 0.2867893$
		PIII3	$\beta=14.10222$, $\xi=43.06666$, $\alpha=20.81451$
		LPIII3	$\beta=401.548819$, $\alpha=85.456119$, $\xi=1.093052$
		PV3	$\alpha=15.42732$, $\gamma=23.36739$, $\beta=4532.59272$
		GUM2	$\beta=299.31065$, $\alpha=68.60258$
		GEV3	$\alpha = 304.2195965$, $\beta= 70.4309848$, $\gamma=0.1310111$
		W2	$\alpha = 4.46416$, $k = 367.60520$

		GP3	$\xi=176.9997592$, $\sigma=279.6301341$, $k= -0.6577304$
11	Gangetic West Bengal	N2	$\mu = 387.54918$, $\sigma = 86.88402$
		LN2	$\mu = 5.9351321$, $\sigma = 0.2225377$
		PIII3	$\beta=10.56873$, $\xi=105.08262$, $\alpha=26.72661$
		LPIII3	$\beta=490.6165071$, $\alpha=99.3735339$, $\xi=0.9980237$
		PV3	$\alpha=17.31328$, $\gamma=26.77757$, $\beta=5895.36219$
		GUM2	$\beta=346.50372$, $\alpha= 74.30961$
		GEV3	$\alpha = 350.938067$, $\beta= 76.165986$, $\gamma=$ 0.111035
		W2	$\alpha = 4.628275$, $k= 422.616657$
		GP3	$\xi=196.2989598$, $\sigma=318.1443294$, $k= -0.7249966$
12	Gujarat	N2	$\mu = 402.9098$, $\sigma = 142.2149$
		LN2	$\mu = 5.9321181$, $\sigma = 0.3760496$
		PIII3	$\beta=7.227182$, $\xi=10.381899$, $\alpha=54.316938$
		LPIII3	$\beta=127.462259$, $\alpha=29.389169$, $\xi=1.594923$
		PV3	$\alpha=6.326561$, $\gamma=10.218984$, $\beta=2134.616764$
		GUM2	$\beta=334.6726$, $\alpha= 123.5828$
		GEV3	$\alpha = 343.4224778$, $\beta= 127.8388941$, $\gamma= 0.1293348$
		W2	$\alpha= 3.048181$, $k= 451.099049$

		GP3	$\xi=128.9$, $\sigma=637.84886$, $k= -0.82112$
13	Harayana, Chandigarh & Delhi	N2	$\mu = 199.38934$, $\sigma = 69.07209$
		LN2	$\mu = 5.2355524$, $\sigma= 0.3490206$
		PIII3	$\beta=5.26204$, $\xi=39.29165$, $\alpha=30.42500$
		LPIII3	$\beta=117.593670$, $\alpha=30.767854$, $\xi=1.413305$
		PV3	$\alpha=7.170697$, $\gamma=10.140825$, $\beta=1181.827430$
		GUM2	$\beta=167.2034$, $\alpha= 56.3727$
		GEV3	$\alpha= 168.71640144$, $\beta=$ 57.25728746 , $\gamma= 0.04973497$
		W2	$\alpha = 3.062312$, $k= 223.155002$
		GP3	$\xi=72.2998610$, $\sigma=199.9373303$, $k= -0.5881551$
14	Himachal Pradesh	N2	$\mu = 320.39098$, $\sigma = 89.08939$
		LN2	$\mu = 5.7311650$, $\sigma = 0.2776595$
		PIII3	$\beta=3.695671$, $\xi=141.095071$, $\alpha=48.515111$
		LPIII3	$\beta=309.4268929$, $\alpha=63.2716057$, $\xi= 0.8407032$
		PV3	$\alpha=11.65512$, $\gamma=17.76044$, $\beta=3234.15650$
		GUM2	$\beta= 278.1371$, $\alpha = 73.4865$
		GEV3	$\alpha = 280.94519972$, $\beta=$ 75.45650185 , $\gamma= 0.07048886$
		W2	$\alpha = 3.858194$, $k=354.175338$

		GP3	$\xi=169.9998139, \sigma=279.1664712, k=-0.7310884$
15	Jammu and Kashmir and Ladakh	N2	$\mu = 241.65328, \sigma = 68.69435$
		LN2	$\mu = 5.4487900, \sigma = 0.2781298$
		PIII3	$\beta=7.749599, \xi=52.983167, \alpha=24.345781$
		LPIII3	$\beta=251.514858, \alpha=56.830782, \xi=1.023061$
		PV3	$\alpha=11.03221, \gamma=15.91529, \beta=2274.54663$
		GUM2	$\beta= 210.1313, \alpha = 56.1056$
		GEV3	$\alpha = 211.73077376, \beta= 56.76380404, \gamma= 0.05315372$
		W2	$\alpha = 3.60891, k= 267.33333$
		GP3	$\xi=99.799867, \sigma=202.076215, k= -0.554414$
16	Jharkhand	N2	$\mu = 375.2730, \sigma = 73.8108$
		LN2	$\mu = 5.9081964, \sigma = 0.1990172$
		PIII3	$\beta=22.99657, \xi=20.57446, \alpha=15.42406$
		LPIII3	$\beta=323.83242, \alpha=89.51799, \xi=2.29063$
		PV3	$\alpha=20.04651, \gamma=30.48105, \beta=6585.38319$
		GUM2	$\beta= 339.8483, \alpha= 68.7186$
		GEV3	$\alpha = 345.1526571, \beta= 68.7551498, \gamma= 0.1466621$
		W2	$\alpha = 5.132317, k= 405.654717$

		GP3	$\xi=170.1997543$, $\sigma=299.9750541$, $k= -0.6058514$
17	Kerala	N2	$\mu = 796.6508$ $\sigma = 177.7553$
		LN2	$\mu = 6.6569385$, $\sigma = 0.2147422$
		PIII3	$\beta=4.230134$, $\xi=427.241811$, $\alpha=87.327839$
		LPIII3	$\beta=34.919633$, $\alpha=27.450117$, $\xi=$ 5.384827
		PV3	$\alpha=14.28636$, $\gamma=152.17439$ $\beta=8571.98131$
		GUM2	$\beta=714.9893$, $\alpha = 142.3932$
		GEV3	$\alpha = 716.73669644$, $\beta=$ 143.46443773 , $\gamma= 0.02244054$
		W2	$\alpha = 4.412367$, $k=868.297256$
		GP3	$\xi=501.5989939$, $\sigma=735.1355182$ $\beta=$, $k= -0.7117387$
18	Konkan and Goa	N2	$\mu = 1138.3975$ $\sigma = 227.5596$
		LN2	$\mu = 7.0166705$, $\sigma = 0.2062439$
		PIII3	$\beta=12.5845793$, $\xi= 0.1026752$, $\alpha=92.2365542$
		LPIII3	$\beta=206.624933$, $\alpha= 68.568799$ $\beta=$, $\xi=4.003275$
		PV3	$\alpha=11.65065$, $\gamma=267.56426$, $\beta=9363.01286$
		GUM2	$\beta= 1026.0966$, $\alpha = 213.4064$
		GEV3	$\alpha = 1049.7932154$, $\beta=$ 219.4464312 , $\gamma= 0.2053415$
		W2	$\alpha = 5.386529$, $k=1231.481767$
		GP3	$\xi=560$, $\sigma=2008.9677$, $k= -1.5368$

19	Lakshadweep	N2	$\mu = 393.35294$ $\sigma = 85.47134$
		LN2	$\mu = 5.9512388$, $\sigma = 0.2173914$
		PIII3	$\beta=11.56274$, $\xi=101.24480$, $\alpha=25.26288$
		LPIII3	$\beta=253.342313$, $\alpha=72.865352$, $\xi=2.474364$
		PV3	$\alpha=7.86958$, $\gamma=130.39022$, $\beta=1835.23262$
		GUM2	$\beta= 352.5413$, $\alpha = 74.4111$
		GEV3	$\alpha = 357.9239154$, $\beta= 76.7086766$, $\gamma=0.1329249$
		W2	$\alpha = 4.7919$, $k= 428.0186$
		GP3	$\xi=227.5996162$, $\sigma=282.1703665$, $k= -0.6441571$
20	Madhya Maharashtra	N2	$\mu = 269.9546$, $\sigma = 60.7953$
		LN2	$\mu = 5.5720804$, $\sigma = 0.2322571$
		PIII3	$\beta=15.06239$, $\xi=29.36203$, $\alpha=15.97378$
		LPIII3	$\beta=360.945531$, $\alpha=81.033030$, $\xi=$ 1.117663
		PV3	$\alpha=11.88519$, $\gamma=37.48777$, $\beta=2571.43266$
		GUM2	$\beta= 241.78512$, $\alpha = 56.82914$
		GEV3	$\alpha = 246.4506078$, $\beta= 57.5834013$, $\gamma= 0.2023431$
		W2	$\alpha= 4.749932$, $k= 294.262141$
		GP3	$\xi=112.3993703$, $\sigma=264.8899699$, $k= -0.8537473$
21	Marathwada	N2	$\mu = 249.5429$, $\sigma = 71.2374$

		LN2	$\mu = 5.4791669, \sigma = 0.2864457$
		PIII3	$\beta=7.396657, \xi=54.461813, \alpha=26.374213$
		LPIII3	$\beta=246.4784193, \alpha=54.5198190, \xi=0.9581861$
		PV3	$\alpha=8.898339, \gamma=28.469806, \beta=1777.754698$
		GUM2	$\beta= 217.51844, \alpha = 60.95829$
		GEV3	$\alpha = 219.03457225, \beta= 61.59312206, \gamma= 0.09222163$
		W2	$\alpha = 3.668416, k= 276.096749$
		GP3	$\xi=115.099769, \sigma=229.462541, k= -0.644255$
22	Nagaland, Manipur, Mizoram and Tripura	N2	$\mu = 504.7877, \sigma = 140.1025$
		LN2	$\mu = 6.192187, \sigma = 0.245772$
		PIII3	$\beta=5.169026, \xi=210.945628, \alpha=56.846672$
		LPIII3	$\beta=49.138601, \alpha=, 28.698951, \xi= 4.479979$
		PV3	$\alpha=9.253391, \gamma=120.375629, \beta= 3185.648425$
		GUM2	$\beta=446.5327, \alpha = 101.6552$
		GEV3	$\alpha = 445.27125126, \beta= 101.10456513, \gamma= -0.02205782$
		W2	$\alpha = 3.287509, k= 556.278391$
		GP3	$\xi=267.0996888, \sigma=309.8363808, k= -0.2537328$
23		N2	$\mu = 186.26148, \sigma = 44.91636$

	North Interior Karnataka	LN2	$\mu = 5.198876, \sigma = 0.237695$
		PIII3	$\beta=7.27706, \xi=64.65770, \alpha=16.71055$
		LPIII3	$\beta=239.340301, \alpha=64.926307, \xi=1.512514$
		PV3	$\alpha=14.08450, \gamma=18.79915, \beta=2196.63424$
		GUM2	$\beta= 165.23084, \alpha = 37.69198$
		GEV3	$\alpha = 166.93698397, \beta= 38.50997853, \gamma=0.08282933$
		W2	$\alpha = 4.240675, k= 203.983631$
		GP3	$\xi=100, \sigma=127.37555, k= -0.55582$
24	Orissa	N2	$\mu = 403.93279, \sigma = 73.21762$
		LN2	$\mu = 5.9851822, \sigma = 0.1787798$
		PIII3	$\beta=8.812105, \xi=185.295380, \alpha=24.811018$
		LPIII3	$\beta=282.423372, \alpha=94.019344, \xi=2.981298$
		PV3	$\alpha=14.59878, \gamma=121.28696, \beta=3854.98858$
		GUM2	$\beta= 369.23621, \alpha = 62.27396$
		GEV3	$\alpha = 372.8650684, \beta= 64.0597977, \gamma= 0.1077639$
		W2	$\alpha = 5.656429, k= 435.034463$
		GP3	$\xi=251.7995636, \sigma=249.7094618, k= -0.6663181$
25	Punjab	N2	$\mu = 210.41885, \sigma = 73.60716$
		LN2	$\mu = 5.2906769, \sigma = 0.3411728$

		PIII3	$\beta=2.978834, \xi=79.662337, \alpha=43.895172$
		LPIII3	$\beta=135.03396, \alpha=33.99527, \xi=1.31851$
		PV3	$\alpha=7.803208, \gamma=11.124380, \beta=1364.221869$
		GUM2	$\beta=176.77714, \alpha=57.41609$
		GEV3	$\alpha=176.16903098, \beta=56.99543115, \gamma=-0.01970015$
		W2	$\alpha=3.006285, k=235.672710$
		GP3	$\xi=94.5, \sigma=198.39601, k=-0.57535$
26	Rayalaseema	N2	$\mu=193.42459, \sigma=55.94332$
		LN2	$\mu=5.2247974, \sigma=0.2824719$
		PIII3	$\beta=4.348007, \xi=75.232439, \alpha=27.183045$
		LPIII3	$\beta=177.352391, \alpha=47.065191, \xi=1.456553$
		PV3	$\alpha=9.022109, \gamma=26.970580, \beta=1343.514530$
		GUM2	$\beta=167.62344, \alpha=45.05825$
		GEV3	$\alpha=168.40156203, \beta=45.51833575, \gamma=0.03176225$
		W2	$\alpha=3.567727, k=214.212976$
		GP3	$\xi=100.1997922, \sigma=140.0122952, k=-0.4782861$
27	Saurashtra, Kutch and Diu	N2	$\mu=242.4484, \sigma=118.6471$
		LN2	$\mu=5.3615861, \sigma=0.5364359$

		PIII3	$\beta=2.7914110, \xi=0.1047605,$ $\alpha=92.2169634$
		LPIII3	$\beta=487.415070, \alpha=40.585273, \xi= -$ 6.651717
		PV3	$\alpha=2.537743, \gamma=8.429011$ $, \beta=417.651381$
		GUM2	$\beta= 187.88360, \alpha=94.60113$
		GEV3	$\alpha = 188.79689895, \beta=$ $95.15173420, \gamma= 0.01794065$
		W2	$\alpha = 2.167287, k= 274.361703$
		GP3	$\xi=33.4, \sigma= 423.87813, k= -$ 0.69883
28	South Interior Karnataka	N2	$\mu =253.33852, \sigma = 56.56391$
		LN2	$\mu = 5.5126488, \sigma = 0.2053639$
		PIII3	$\beta=5.070119, \xi=132.183844,$ $\alpha=23.895837$
		LPIII3	$\beta=364.134229, \alpha=93.759802, \xi=$ 1.629045
		PV3	$\alpha=16.39701, \gamma=48.28942$ $, \beta=3152.81054$
		GUM2	$\beta=228.79447, \alpha =42.00173$
		GEV3	$\alpha= 228.08994337, \beta=41.61727912,$ $\gamma= -0.03086515$
		W2	$\alpha = 4.187799, k= 276.220769$
		GP3	$\xi=148, \sigma=139.1465, k= -0.3754$
29	Sub-Himalayan West Bengal and Sikkim	N2	$\mu =722.2811, \sigma = 131.3868$
		LN2	$\mu = 6.5653792, \sigma = 0.1866987$

		PIII3	$\beta=28.739613$, $\xi=9.198317$, $\alpha=24.811883$
		LPIII3	$\beta=191.848194$, $\alpha=72.927113$, $\xi=3.934697$
		PV3	$\alpha=11.2312$, $\gamma=218.4145$ $\beta=5213.6313$
		GUM2	$\beta= 657.4939$, $\alpha =124.6137$
		GEV3	$\alpha = 673.2703747$, $\beta= 127.8875276$, $\gamma= 0.2364892$
		W2	$\alpha = 5.909666$, $k= 777.243259$
		GP3	$\xi=396.0982592$, $\sigma= 589.8963475$ $k= -0.8812634$
30	Tamil Nadu and Pondicherry	N2	$\mu =229.8443$, $\sigma = 68.9699$
		LN2	$\mu = 5.391752$, $\sigma = 0.307243$
		PIII3	$\beta=7.936215$, $\xi=32.288101$, $\alpha=24.899116$
		LPIII3	$\beta=187.402535$, $\alpha=43.970290$, $\xi=1.129441$
		PV3	$\alpha=8.747932$, $\gamma=12.634296$, $\beta=1702.330302$
		GUM2	$\beta= 197.33637$, $\alpha =59.65409$
		GEV3	$\alpha = 200.974802$, $\beta= 60.852047$, $\gamma=$ 0.113191
		W2	$\alpha = 3.495574$, $k= 254.979269$
		GP3	$\xi=85.9998900$, $\sigma= 249.8834842$, $k= -0.7075765$
31	Telangana	N2	$\mu = 296.44508$, $\sigma = 84.68354$
		LN2	$\mu = 5.6513593$, $\sigma = 0.2883285$

		PIII3	$\beta=10.42843, \xi=23.93303$, $\alpha=26.13295$
		LPIII3	$\beta=276.1458652$, $\alpha=57.0625966$, $\xi=0.8118558$
		PV3	$\alpha=10.10949$, $\gamma=15.26585$, $\beta=2583.16735$
		GUM2	$\beta=256.92910$, $\alpha=73.21928$
		GEV3	$\alpha=260.59450493$, $\beta=$ 74.14287212 , $\gamma=0.09182375$
		W2	$\alpha=3.55359$, $k=327.64815$
		GP3	$\xi=106.0995045$, $\sigma=294.5240749$, $k=-0.5251243$
32	Uttaranchal	N2	$\mu=442.27377$, $\sigma=86.18858$
		LN2	$\mu=6.0721638$, $\sigma=0.2021041$
		PIII3	$\beta=23.46367$, $\xi=16.72667$, $\alpha=18.13662$
		LPIII3	$\beta=194.533601$, $\alpha=67.495803$, $\xi=$ 3.189974
		PV3	$\alpha=11.2154$, $\gamma=103.0434$, $\beta=3510.6092$
		GUM2	$\beta=399.81539$, $\alpha=83.53344$
		GEV3	$\alpha=409.7453301$, $\beta=84.0525553$, $\gamma=0.2255856$
		W2	$\alpha=5.475439$, $k=477.729365$
		GP3	$\xi=206.5995177$, $\sigma=409.9551242$, $k=-0.8547965$
33	Vidarbha	N2	$\mu=364.08607$, $\sigma=78.22542$
		LN2	$\mu=5.8747505$, $\sigma=0.2127634$

		PIII3	$\beta=9.632723, \xi=120.733395,$ $\alpha=25.263112$
		LPIII3	$\beta=524.1833193, \alpha=107.5058008$ $, \xi=0.9988823$
		PV3	$\alpha=11.05163, \gamma=94.84639,$ $\beta=2723.90005$
		GUM2	$\beta=327.10325, \alpha=66.76114$
		GEV3	$\alpha=331.0412825, \beta=68.5392663,$ $\gamma=0.1088603$
		W2	$\alpha=4.802761, k=396.004564$
		GP3	$\xi=196.8996505, \sigma=299.8769519,$ $k=-0.7570095$
34	West Madhya Pradesh	N2	$\mu=362.16311, \sigma=79.1110$
		LN2	$\mu=5.8684287, \sigma=0.2183062$
		PIII3	$\beta=13.08642, \xi=75.73668,$ $\alpha=21.88729$
		LPIII3	$\beta=501.8544416, \alpha=102.3422920,$ $\xi=0.9647195$
		PV3	$\alpha=14.95319, \gamma=49.29852,$ $\beta=4379.96465$
		GUM2	$\beta=324.53339, \alpha=69.08733$
		GEV3	$\alpha=329.6017076, \beta=70.9330038,$ $\gamma=0.1359603$
		W2	$\alpha=4.762451, k=394.261877$
		GP3	$\xi=202.899451, \sigma=279.992122, k=-0.773741$
35	West Rajasthan	N2	$\mu=133.53689, \sigma=56.97289$
		LN2	$\mu=4.7958749, \sigma=0.4653585$

		PIII3	$\beta = 1.387277$, $\xi = 26.717535$, $\alpha = 100.361731$
		LPIII3	$\beta = 291.053456$, $\alpha = 35.731522$, $\xi = -3.354381$
		PV3	$\alpha = 28.68228$, $\gamma = -161.95325$, $\beta = 8180.21567$
		GUM2	$\beta = 106.91306$, $\alpha = 47.67441$
		GEV3	$\alpha = 108.94134200$, $\beta = 48.62847152$, $\gamma = 0.07814724$
		W2	$\alpha = 2.487643$, $k = 150.644035$
		GP3	$\xi = 26.8$, $\sigma = 148.92138$, $k = -0.47923$
36	West Uttar Pradesh	N2	$\mu = 300.92295$, $\sigma = 75.55604$
		LN2	$\mu = 5.6712643$, $\sigma = 0.2773545$
		PIII3	$\beta = 11.77669$, $\xi = 23.54019$, $\alpha = 23.55781$
		LPIII3	$\beta = 304.4255730$, $\alpha = 61.7318914$, $\xi = 0.7395887$
		PV3	$\alpha = 8.772593$, $\gamma = 29.275376$, $\beta = 2149.629899$
		GUM2	$\beta = 262.66203$, $\alpha = 75.15246$
		GEV3	$\alpha = 273.9280963$, $\beta = 76.4852498$, $\gamma = -0.2798968$
		W2	$\alpha = 4.50435$, $k = 329.77330$
		GP3	$\xi = 124.1997344$, $\sigma = 328.7846485$, $k = -0.8514325$

4.1.1. Selecting the best-fit Results

Our study's collected data revealed that each region's rainfall rang differed significantly. The variations in rainfall can be attributed in part to the site's geographic position including its longitude latitude evaluation and surrounding environmental conditions. Arunachal Pradesh has the highest amount of precipitation of 2362.8mm while West Rajasthan has the lowest amount of Precipitation of 26.9mm.

Table 2 displays the Regions names, the findings of the statistical parameters mean, standard deviation, coefficient of skewness, minimum value, maximum value, medium and skewness for each of the regions that we've employed in our study. The absolute value of skewness reveals the relative magnitude of the deviations from the mean. West Rajasthan has a mean score of 133.5369mm, the lowest figure, whereas Coastal Karnataka records a mean value of 1180.852mm, the highest figure. The average for 14 regions lies within 0-300mm, for 16 regions it lies within 300mm-600mm, for 4 regions it lies within 600mm-900mm, for 2 regions lies it within 900mm-1200mm. Arunachal Pradesh records the highest scattered rainfall of 362.9061mm, while North Inter Karnataka records the lowest scattered rainfall of 45.10158mm. The standard deviation for 33 regions falls between 0 - 200 mm, while for the other 3 regions it falls between 200mm-400mm. The amount of standard deviation with respect to its mean is shown by the coefficient of variation (CV), a relative measure of variability. The CV in Saurashtra, Kutch and Diu is 49.13885 at the maximum, while it is 17.0942 in Chattisgarh, the minimum value. The coefficient to variation of all regions lies within 0-50mm. The coefficient of skewness of 4 stations are greater than 1, thus can be regarded as slightly skewed. 20 stations lie with the range of 0.5 to 1, thus highly skewed , while 12 stations are approximately symmetric, lying with the range of -0.5 to 0.5.

Table 2: Regions with their Statistical Summary

S.NO.	Region Names	Mean	SD	CV(%)	Min	Max	Median	Skewness
1	Andaman and Nicobar Island	604.4552	131.4605	21.7486	392	973.1	567.8	0.7674096
2	Arunachal Pradesh	802.3817	362.9061	45.22861	302.6	2362.8	681.45	1.212462
3	Assam and Meghalaya	577.7328	107.1159	18.54073	355.8	995.2	569.8	0.8440902
4	Bihar	369.559	78.66892	21.28724	191.2	580.1	363.75	0.104587
5	Chattisgarh	440.7197	75.33749	17.0942	240.8	630.6	437.65	0.05875039
6	Coastal Andhra Pradesh	259.8246	67.02783	25.79734	149.6	527.2	251.55	1.224076
7	Coastal Karnataka	1180.852	244.5751	20.71174	699.5	1904.9	1135	0.6774014
8	East Madhya Pradesh	432.8246	97.33554	22.48845	205.9	713	429.55	0.1933738
9	East Rajasthan	278.7172	72.76117	26.10573	108.8	440.6	281.75	- 0.03764708
10	East Uttar Pradesh	336.5975	78.39975	23.29184	177	598.8	335.15	0.5334382
11	Gangetic West Bengal	387.5492	87.24231	22.51129	196.3	633.1	381.65	0.5766691
12	Gujarat	402.9098	142.8014	35.44252	128.9	903.3	388.65	0.4558145
13	Haryana, Chandigarh & Delhi	199.3893	69.35692	34.78467	72.3	405.3	189.45	0.6989948
14	Himachal Pradesh	320.391	89.45677	27.92113	170	548.7	301.8	0.4890317

15	Jammu and Kashmir and Ladakh	241.6533	68.97763	28.54405	99.8	454.6	230	0.8752227
16	Jharkhand	375.273	74.11518	19.74967	170.2	660.2	370.2	0.506226
17	Kerala	796.6508	178.4883	22.40483	501.6	1526.5	768.65	0.9400015
18	Konkan and Goa	1138.398	228.498	20.0719	578.1	1867.2	1119.15	0.1596355
19	Lakshadweep	393.3529	85.83274	21.8208	227.6	660.8	381.2	0.4974212
20	Madhya Maharashtra	271.6107	61.45874	22.62751	112.4	421.7	264.95	0.2648394
21	Marathwada	251.5254	72.36116	28.76893	115.1	467.9	246.1	0.6100446
22	Nagaland, Manipur, Mizoram and Tripura	504.7877	140.6802	27.86919	267.1	1347.2	483.3	2.312788
23	North Interior Karnataka	186.2615	45.10158	24.21412	104	323.1	184.95	0.7147661
24	Orissa	403.9328	73.51955	18.20094	251.8	624.9	398.2	0.5450739
25	Punjab	210.4189	73.9107	35.12551	95	434.1	201.15	0.8625917
26	Rayalaseema	193.4246	56.17402	29.04182	100.2	383.8	183.85	0.8355458
27	Saurashtra, Kutch and Diu	242.4484	119.1363	49.13885	33.5	634	224.85	0.8395147
28	South Interior Karnataka	253.3385	56.79716	22.41947	148.4	492.7	245.75	1.498194
29	Sub-Himalayan West Bengal and Sikkim	722.2811	131.9286	18.26555	396.1	1064.6	731.65	0.160282
30	Tamil Nadu and Pondicherry	229.8443	69.25431	30.13097	86	436.1	225.4	0.6403528

31	Telangana	296.4451	85.03275	28.68415	106.1	656.2	289.2	0.8603693
32	Uttaranchal	442.2738	86.544	19.56797	206.6	685.2	441.9	0.1815806
33	Vidarbha	364.0861	78.548	21.57402	196.9	591	360.75	0.5834892
34	West Madhya Pradesh	362.1631	79.43725	21.93411	202.9	561.6	349.85	0.5243684
35	West Rajasthan	133.5369	57.20783	42.84047	26.9	325.3	124.1	0.732636
36	West Uttar Pradesh	300.923	75.86761	25.21164	124.2	509.6	307.8	-0.132876

The Regions name, best fit test statistic results of K-S, A-D, CVM and BIC are tabulated in the Table 3. Using several goodness of fit tests including K-S, AD, BIC and CVM tests we have computed a test statistic for each of the 36 regions. The best fit result is taken as the smallest goodness of fit results among all the probability distributions developed for each region. For e.g. in Bihar , GEV3 distribution by CVM test gave the best fit, in Coastal Karnataka PV3 distribution by K-S test and GEV3 distribution by A-D test gave the best fits. In Gangetic-West Bengal LPIII3 distribution by K-S test and GEV3 distribution by BIC test gave the best fit results. In Gujarat PIII3 distribution by K-S test and GEV3 distribution by CVM test gave the best fit results.

Table 3. Regions along with their Goodness of fit test Results

S. NO.	REGION NAME	BEST-FIT TEST STATISTIC RESULTS			
		By K-S test	By A-D test	By CVM test	By BIC
1	Andaman and Nicobar Island	PIII3(0.0460335)	PIII3(0.2618804)	PIII3(0.04594755)	GEV3(1451.585)
2	Arunachal Pradesh	PIII3(0.0592176)	PIII3(0.5487525)	PIII3(0.09042643)	PIII3(1497.571)
3	Assam and Meghalaya	GUM2(0.04969022)	LPIII3(0.4233746)	GEV3(0.05805063)	GEV3(1484.181)
4	Bihar	GEV3(0.03364188)	GEV3(0.2038991)	GEV3(0.02881128)	GEV3(1419.113)
5	Chattisgarh	PIII3(0.0272884)	GEV3(0.1563358)	PIII3(0.02308132)	GEV3(1408.831)
6	Coastal Andhra Pradesh	LPIII3(0.03233729)	LPIII3(0.1160044)	LPIII3(0.01601914)	GEV3(1354.413)
7	Coastal Karnataka	PV3(0.04568268)	GEV3(0.2131978)	PV3(0.02703539)	GEV3(1686.102)
8	East Madhya Pradesh	PIII3(0.0323409)	GEV3(0.1577437)	GEV3(0.02229756)	GEV3(1470.868)
9	East Rajasthan	N2(0.05978924)	N2(0.4742756)	N2(0.07459674)	GEV3(1399.879)
10	East Uttar Pradesh	PIII3(0.03145538)	PIII3(0.1700223)	PIII3(0.02632736)	GEV3(1413.74)
11	Gangetic West Bengal	LPIII3((0.03486673)	LPIII3((0.2700721)	LPIII3(0.04130295)	GEV3(1436.945)

12	Gujarat	PIII3(0.03875574)	GEV3(0.2393991)	GEV3(0.03172663)	GEV3(1560.246)
13	Harayana, Chandigarh & Delhi	PV3(0.03877243)	PIII3(0.230268)	LN2(0.03356645)	GEV3(1376.273)
14	Himachal Pradesh	PIII3(0.04811033)	PIII3(0.4342019)	PIII3(0.06070517)	LN2(1441.582)
15	Jammu and Kashmir and Ladakh	LPIII3(0.06028093)	GUM2(0.3241459)	LPIII3(0.05060151)	GEV3(1372.594)
16	Jharkhand	PIII3(0.04337345)	PIII3(0.3685537)	PIII3(0.05182156)	GEV3(1403.182)
17	Kerala	PIII3(0.03325194)	PIII3(0.2060719)	PIII3(0.02648761)	GEV3(1603.331)
18	Konkan and Goa	N2(0.05163159)	N2(0.4005455)	GEV3(0.05882603)	GEV3(1679.907)
19	Lakshadweep	GEV3((0.03508986)	GEV3((0.1574036)	GEV3(0.0218735)	GEV3(1400.141)
20	Madhya Maharashtra	GEV3(0.03763848)	GEV3(0.1985549)	GEV3(0.02765042)	GEV3(1322.398)
21	Marathwada	GEV3(0.05121493)	GEV3(0.2694213)	GEV3(0.04252156)	GEV3(1353.484)
22	Nagaland, Manipur, Mizoram and Tripura	GUM2(0.07002081)	GUM2(0.6035473)	GUM2(0.09810056)	GEV3(1520.869)

23	North Interior Karnataka	PIII3(0.04306996)	LN2(0.3494355)	LN2(0.05334883)	LN2(1273.784)
24	Orissa	GUM2(0.04020205)	PIII3(0.1954042)	PIII3(0.03447055)	GEV3(1395.424)
25	Punjab	GEV3(0.04121634)	LPIII3(0.3276462)	LN2(0.04545807)	LN2(1384.365)
26	Rayalaseema	GEV3(0.03838296)	LN2(0.1945855)	LN2(0.02621052)	LN2(1322.221)
27	Saurashtra, Kutch and Diu	GUM2(0.0377223)	GUM2(0.2443722)	GUM2(0.03392444)	GEV3(1504.4)
28	South Interior Karnataka	GUM2(0.04558367)	GEV3(0.2772964)	GUM2(0.03575529)	GEV3(1307.892)
29	Sub- Himalayan West Bengal and Sikkim	GEV3(0.04836635)	GEV3(0.1871729)	GEV3(0.03658562)	GEV3(1545.061)
30	Tamil Nadu and Pondicherry	PIII3(0.03609871)	GEV3(0.2639394)	GEV3(0.02684715)	GEV3(1380.875)
31	Telangana	PIII3(0.06358048)	LN2(0.3565565)	LN2(0.06325132)	GEV3(1429.916)
32	Uttaranchal	PIII3(0.03859548)	PIII3(0.1627497)	PIII3(0.02674931)	GEV3(1443.045)
33	Vidarbha	LPIII3(0.04553887)	PIII3(0.4031836)	LN2(0.06284046)	GEV3(1411.383)

34	West Madhya Pradesh	LPIII3(0.04691905)	LPIII3(0.3249219)	LPIII3(0.04308023)	GEV3(1415.838)
35	West Rajasthan	GEV3(0.03290925)	PV3(0.1493329)	GEV3(0.02214043)	GEV3(1331.375)
36	West Uttar Pradesh	W2(0.04560779)	W2(0.3041559)	W2(0.05069626)	W2(1409.902)

In order to determine which distribution is best suited for each region, a ranking system was created, with rank 1 being the best rank, rank 2 representing the second best and so on as shown in Table 4. Table 4 displays regions names, distributions names, goodness of fit such as KS, AD, CVM AND BIC and their rankings and total rank is the sum of rank of each distribution. The distribution type with the lowest sum of the four rankings is selected as a best fit distribution for each of the stations. Seven stations had two distributions with the same value of best score (a tie between two types). They were LN2 and LPIII3 distributions in Gangetic West Bengal, LPIII3 and GUM2 distributions in Jammu and Kashmir and Ladakh, PIII3 and GEV3 distributions in Jharkhand, N2 and GEV3 distributions in Konkan and Goa, PIII3 and GEV3 distributions in Orissa, LN2 and GEV3 distributions in Rayalaseema, GEV3 and N2 distributions in Uttaranchal. Some examples of the distributions with their lowest ranks have been given. For e.g. In Arunachal Pradesh the lowest total rank is 4, shown by PIII3 distribution, therefore we conclude that PIII3 is the best fit distribution for Arunachal Pradesh. In Assam and Meghalaya, the lowest rank is 6, shown by the GEV3 distribution, in Bihar the lowest rank is 4 shown by the GEV3 distribution, in Coastal Karnataka is 8 shown by GEV3 distribution. In Gujarat the lowest rank is 5 shown by GEV3 distribution. Table 4 show that just 2 regions i.e. 13 and 33 shower distinct distribution throughout all the 4 goodness of a test. Where seven regions are best fitted by the same distribution over all four goodness affect tests.

Table 4: Regions, their distributions and Ranks of Goodness-of-fit

S. No	Region Names	Distribution Name	Rank KS	Rank AD	Rank CVM	Rank BIC	Total Rank
1	Andaman and Nicobar Island	N2	9	7	8	8	32
		LN2	7	6	6	5	24
		PIII3	1	1	1	3	6
		LPIII3	2	2	2	4	10
		PV3	3	3	3	6	15
		GUM2	5	5	5	2	17
		GEV3	4	4	4	1	13
		W2	8	9	9	9	35
		GP3	6	8	7	7	28
2	Arunachal Pradesh	N2	9	9	9	9	36
		LN2	5	5	5	3	18
		PIII3	1	1	1	1	4
		LPIII3	2	2	2	4	10
		PV3	3	3	3	5	14
		GUM2	6	6	7	6	25
		GEV3	4	4	4	2	14
		W2	8	8	8	8	32
		GP3	7	8	6	7	28
3	Assam and Meghalaya	N2	7	7	7	6	27
		LN2	6	3	4	2	15
		PIII3	5	4	3	5	17
		LPIII3	3	1	2	4	10
		PV3	4	6	6	7	23
		GUM2	1	5	5	3	14
		GEV3	2	2	1	1	6
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
4	Bihar	N2	3	3	3	2	11

		LN2	4	5	4	5	18
		PIII3	2	2	2	4	10
		LPIII3	5	6	6	6	23
		PV3	8	8	8	8	32
		GUM2	7	7	7	7	28
		GEV3	1	1	1	1	4
		W2	6	4	5	3	18
		GP3	9	9	9	9	36
5	Chattisgarh	N2	5	2	3	2	12
		LN2	2	4	4	3	13
		PIII3	1	3	1	5	10
		LPIII3	3	5	5	6	19
		PV3	7	8	8	8	31
		GUM2	6	7	7	7	27
		GEV3	4	1	2	1	8
		W2	8	6	6	4	24
6	Coastal Andhra Pradesh	GP3	9	9	9	9	36
		N2	7	7	7	7	28
		LN2	6	6	6	3	21
		PIII3	2	5	2	5	14
		LPIII3	1	1	1	4	7
		PV3	4	3	3	6	16
		GUM2	5	4	5	2	16
		GEV3	3	2	4	1	10
7	Coastal Karnataka	W2	8	8	8	9	33
		GP3	9	9	9	8	35
		N2	8	7	7	7	29
		LN2	6	6	6	2	20
		PIII3	4	5	4	4	17
		LPIII3	5	4	5	5	19
		PV3	1	2	1	6	10

		GUM2	2	3	2	3	10
		GEV3	3	1	3	1	8
		W2	7	8	8	8	31
		GP3	9	9	9	9	36
8	East Madhya Pradesh	N2	6	3	3	2	14
		LN2	3	4	4	3	14
		PIII3	1	2	2	5	10
		LPIII3	4	6	5	7	22
		PV3	8	8	8	8	32
		GUM2	7	7	7	6	27
		GEV3	2	1	1	1	5
		W2	5	5	6	4	20
		GP3	9	9	9	9	36
9	East Rajasthan	N2	1	1	1	3	6
		LN2	5	5	5	5	20
		PIII3	4	4	4	4	16
		LPIII3	7	7	7	7	28
		PV3	8	8	8	8	32
		GUM2	6	6	6	6	24
		GEV3	2	2	2	1	7
		W2	3	3	3	2	11
		GP3	9	9	9	9	36
10	East Uttar Pradesh	N2	6	5	4	4	19
		LN2	3	3	3	7	16
		PIII3	1	1	1	3	6
		LPIII3	4	4	5	5	18
		PV3	7	7	7	6	27
		GUM2	5	6	6	2	19
		GEV3	2	2	2	1	7
		W2	8	8	8	8	32
		GP3	9	9	9	9	36

11	Gangetic West Bengal	N2	7	7	7	7	28
		LN2	2	2	2	2	8
		PIII3	3	4	3	4	14
		LPIII3	1	1	1	5	8
		PV3	6	5	5	6	22
		GUM2	5	6	6	3	20
		GEV3	4	3	4	1	12
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
12	Gujarat	N2	6	4	5	6	21
		LN2	5	6	6	4	21
		PIII3	1	2	3	5	11
		LPIII3	7	7	7	7	28
		PV3	8	8	8	8	32
		GUM2	4	5	4	3	16
		GEV3	2	1	1	1	5
		W2	3	3	2	2	10
		GP3	9	9	9	9	36
13	Haryana, Chandigarh & Delhi	N2	8	8	8	8	32
		LN2	3	3	1	2	9
		PIII3	5	1	3	4	13
		LPIII3	2	5	4	5	16
		PV3	1	6	6	7	20
		GUM2	4	2	2	3	11
		GEV3	6	4	5	1	16
		W2	7	7	7	6	27
		GP3	9	9	9	9	36
14	Himachal Pradesh	N2	8	7	8	8	31
		LN2	3	4	5	1	13
		PIII3	1	1	1	4	7
		LPIII3	2	2	3	6	13

		PV3	6	5	4	7	22
		GUM2	5	3	2	3	13
		GEV3	4	6	6	2	18
		W2	9	8	7	9	33
		GP3	7	9	9	5	30
15	Jammu and Kashmir and Ladakh	N2	7	7	7	7	28
		LN2	4	4	4	2	14
		PIII3	6	6	6	5	23
		LPIII3	1	2	1	4	8
		PV3	5	5	5	6	21
		GUM2	2	1	2	3	8
		GEV3	3	3	3	1	10
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
16	Jharkhand	N2	3	4	3	3	13
		LN2	4	3	4	2	13
		PIII3	1	1	1	4	7
		LPIII3	5	5	5	5	20
		PV3	6	6	6	7	25
		GUM2	7	7	7	6	27
		GEV3	2	2	2	1	7
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
17	Kerala	N2	7	7	7	6	27
		LN2	6	6	6	8	26
		PIII3	1	1	1	3	6
		LPIII3	2	2	2	4	10
		PV3	5	5	5	5	20
		GUM2	4	4	4	2	14
		GEV3	3	3	3	1	10
		W2	8	8	8	7	31

		GP3	9	9	9	9	36
18	Konkan and Goa	N2	1	1	2	2	6
		LN2	3	4	4	3	14
		PIII3	8	8	8	8	32
		LPIII3	4	5	5	5	19
		PV3	7	7	7	7	28
		GUM2	6	6	6	6	24
		GEV3	2	2	1	1	6
		W2	5	3	3	4	15
		GP3	9	9	9	9	36
19	Lakshadweep	N2	6	6	6	6	24
		LN2	2	3	3	2	10
		PIII3	3	2	2	4	11
		LPIII3	4	4	4	5	17
		PV3	8	8	8	8	32
		GUM2	5	5	5	3	18
		GEV3	1	1	1	1	4
		W2	7	7	7	7	28
		GP3	9	9	9	9	36
20	Madhya Maharashtra	N2	4	4	3	2	13
		LN2	3	3	4	3	13
		PIII3	2	2	2	4	10
		LPIII3	5	5	5	6	21
		PV3	8	8	8	9	33
		GUM2	6	6	6	7	25
		GEV3	1	1	1	1	4
		W2	7	7	7	5	26
		GP3	9	9	9	8	35
21	Marathwada	N2	6	6	6	5	23
		LN2	3	3	3	2	11
		PIII3	2	2	2	3	9

		LPIII3	5	4	5	4	18
		PV3	8	7	7	9	31
		GUM2	4	5	4	7	20
		GEV3	1	1	1	1	4
		W2	7	8	8	6	29
		GP3	9	9	9	7	34
22	Nagaland, Manipur, Mizoram and Tripura	N2	7	7	7	7	28
		LN2	2	4	4	3	13
		PIII3	3	5	5	6	19
		LPIII3	4	2	2	4	12
		PV3	6	6	6	5	23
		GUM2	1	1	1	2	5
		GEV3	5	3	3	1	12
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
23	North Interior Karnataka	N2	7	7	7	7	28
		LN2	2	1	1	1	5
		PIII3	1	3	2	4	10
		LPIII3	4	4	4	5	17
		PV3	6	6	6	6	24
		GUM2	5	5	5	3	18
		GEV3	3	2	3	2	10
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
24	Orissa	N2	7	7	8	7	29
		LN2	6	5	6	2	19
		PIII3	3	1	1	4	9
		LPIII3	5	3	3	5	16
		PV3	2	6	5	6	19
		GUM2	1	4	4	3	12
		GEV3	4	2	2	1	9

		W2	8	8	7	8	31
		GP3	9	9	9	9	36
25	Punjab	N2	8	8	8	9	33
		LN2	3	3	1	1	8
		PIII3	5	2	5	4	16
		LPIII3	4	1	4	5	14
		PV3	6	6	6	6	24
		GUM2	2	5	2	3	12
		GEV3	1	4	3	2	10
		W2	7	7	7	8	29
		GP3	9	9	9	7	34
26	Rayalaseema	N2	7	7	7	7	28
		LN2	4	1	1	1	7
		PIII3	5	5	4	4	18
		LPIII3	2	3	3	5	13
		PV3	6	6	6	6	24
		GUM2	3	4	5	3	15
		GEV3	1	2	2	2	7
		W2	8	8	8	9	33
		GP3	9	9	9	8	35
27	Saurashtra, Kutch and Diu	N2	8	7	7	7	29
		LN2	3	3	3	4	13
		PIII3	6	6	6	5	23
		LPIII3	4	5	4	6	19
		PV3	7	8	8	9	32
		GUM2	1	1	1	2	5
		GEV3	2	2	2	1	7
		W2	5	4	5	3	17
		GP3	9	9	9	8	35
28	South Interior Karnataka	N2	7	7	7	7	28
		LN2	6	6	6	4	22

		PIII3	5	4	4	5	18
		LPIII3	4	5	5	6	20
		PV3	3	3	2	3	11
		GUM2	1	2	1	2	6
		GEV3	2	1	3	1	7
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
29	Sub-Himalayan West Bengal and Sikkim	N2	2	3	3	2	10
		LN2	4	4	4	3	15
		PIII3	3	2	2	4	11
		LPIII3	5	5	5	6	21
		PV3	8	8	8	8	32
		GUM2	6	7	7	7	27
		GEV3	1	1	1	1	4
		W2	7	6	6	5	24
		GP3	9	9	9	9	36
30	Tamil Nadu and Pondicherry	N2	6	6	6	4	22
		LN2	8	3	4	9	24
		PIII3	1	2	2	3	8
		LPIII3	4	5	5	5	19
		PV3	5	7	7	7	26
		GUM2	3	4	3	2	12
		GEV3	2	1	1	1	5
		W2	7	8	8	6	29
		GP3	9	9	9	8	35
31	Telangana	N2	5	6	6	6	23
		LN2	3	1	1	2	7
		PIII3	1	2	2	4	9
		LPIII3	7	4	4	5	20
		PV3	8	7	7	7	29
		GUM2	6	5	5	3	19

		GEV3	2	3	3	1	9
		W2	4	8	8	8	28
		GP3	9	9	9	9	36
32	Uttaranchal	N2	2	2	1	2	7
		LN2	4	4	4	3	15
		PIII3	1	3	3	4	11
		LPIII3	5	5	5	6	21
		PV3	8	8	8	8	32
		GUM2	6	7	7	7	27
		GEV3	3	1	2	1	7
		W2	7	6	6	5	24
		GP3	9	9	9	9	36
33	Vidarbha	N2	6	7	6	6	25
		LN2	2	2	1	2	7
		PIII3	4	1	2	4	11
		LPIII3	1	4	4	5	14
		PV3	7	6	7	7	27
		GUM2	5	5	5	3	18
		GEV3	3	3	3	1	10
		W2	8	8	8	8	32
		GP3	9	9	9	9	36
34	West Madhya Pradesh	N2	7	7	7	5	26
		LN2	6	4	4	9	23
		PIII3	2	3	3	3	11
		LPIII3	1	1	1	4	7
		PV3	5	6	6	6	23
		GUM2	4	5	5	2	16
		GEV3	3	2	2	1	8
		W2	8	8	8	7	31
		GP3	9	9	9	8	35
35		N2	6	6	6	5	23

	West Rajasthan	LN2	5	4	4	9	22
		PIII3	9	9	9	8	35
		LPIII3	7	7	7	6	27
		PV3	2	1	2	4	9
		GUM2	3	3	3	2	11
		GEV3	1	2	1	1	5
		W2	4	5	5	3	17
		GP3	8	8	8	7	31
36	West Uttar Pradesh	N2	3	4	4	2	13
		LN2	4	2	2	9	17
		PIII3	5	5	5	4	19
		LPIII3	6	6	7	6	25
		PV3	8	8	8	7	31
		GUM2	7	7	6	5	25
		GEV3	2	3	3	3	11
		W2	1	1	1	1	4
		GP3	9	9	9	8	35

The regions names and distribution of lowest rank sums are displayed in Table 5. This table indicates the best fit distribution according to the lowest rank sum. For e.g.; - In Arunachal Pradesh, PIII3 is the best-fitted distribution with 4 as the rank sum, in Assam and Meghalaya GEV3 distribution is the best-fitted with 6 as the lowest rank sum. Similarly, we have displayed the best fit distribution for all regions. From table 2 and table 5 we can say that, Among the lowest ranked distribution, all the regions with mean maximum monthly rainfall greater than 400 mm is best-fitted by either GEV3 or PIII3 distribution except region no. 22 which is best-fit by GUM2 distribution. The 2 regions with negative skewness, region no. 9, with N2 distribution and region no. 36 with W2 distribution, are best fitted by two parameter distributions. All regions with skewness between 0 and 0.5 are symmetric and best fitted by PIII3 distribution except region number 14. GEV3 is the highest ranked-distribution in these regions. The regions with highly skewed distributions, best fitted by PIII3 and LPIII3 have skewness between 0.5-1. Among all the distributions it was found that in the best score result, 47% was best-fitted by GEV3 distribution, 20% was best fitted by each PIII3 and LN2

distributions and 11% were best fitted by LPIII3 distribution and GUM2 , N2 and W2 distributions gave some best-fit results of 2% each. However, GP3 didn't gave any best-fit results. Based on Table 4 and 5, we can say that while PV3 is not among the lowest-ranking distributions in any region, it did provide some best fit results in areas such as Coastal Karnataka (shown by K-S and CVM test), Haryana, Chandigarh and Delhi (shown by K-S test) and West Rajasthan (shown by A-D test).

(Table 5. Shows the regions along with their distributions that have the lowest rank sums). For e.g.: - region no. 5 is Chattisgarh, which has GEV3 as the best probability distribution and has smallest sum of ranks from all four Goodness-of-fit tests, i.e. 8.

Table 5: Regions with their distributions of lowest rank sums

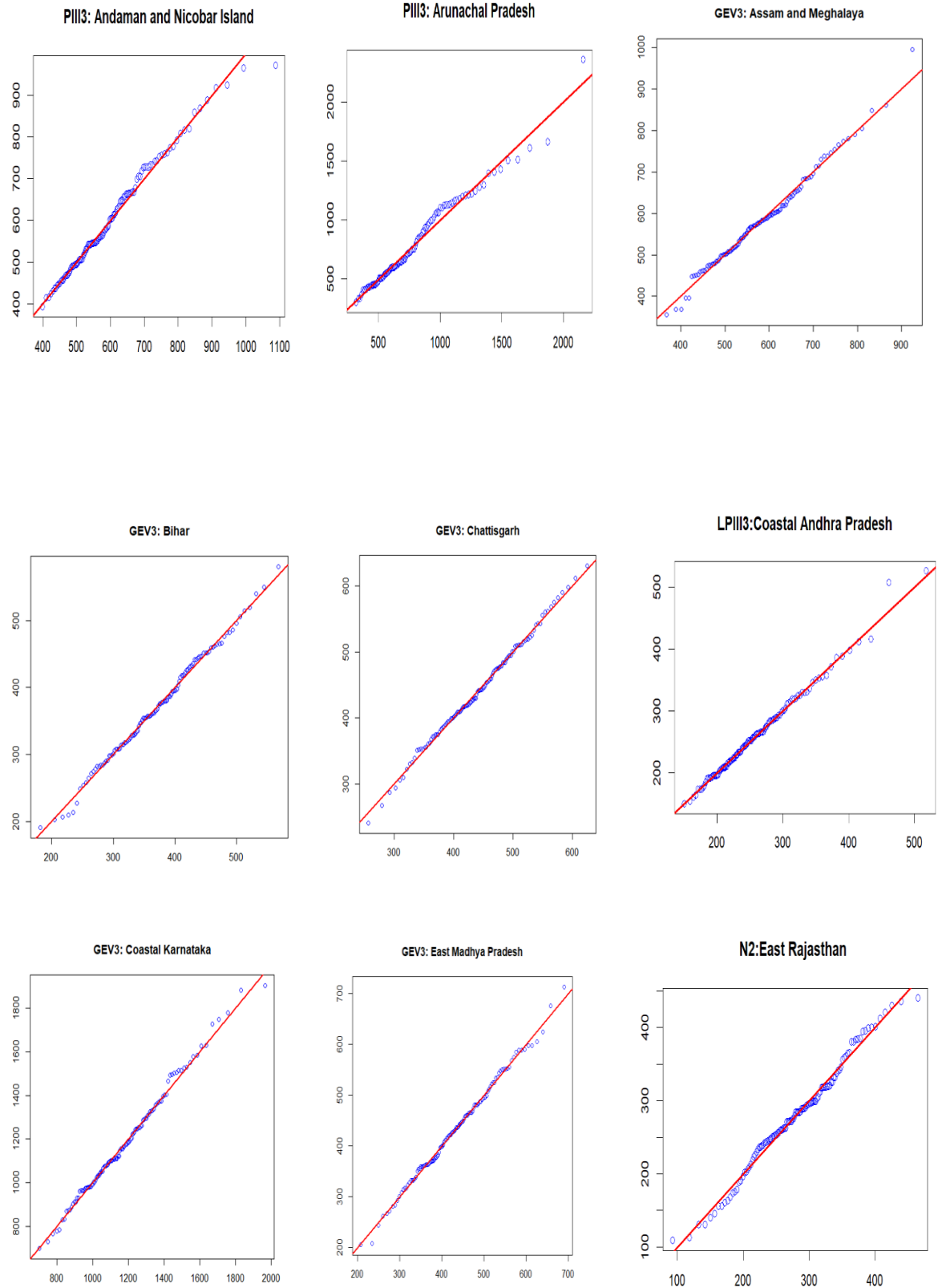
S. NO.	REGION NAME	Lowest Ranked Distribution (Sum of Ranks)
1	Andaman and Nicobar Island	PIII3(6)
2	Arunachal Pradesh	PIII3(4)
3	Assam and Meghalaya	GEV3(6)
4	Bihar	GEV3(4)
5	Chattisgarh	GEV3(8)
6	Coastal Andhra Pradesh	LPIII3(7)
7	Coastal Karnataka	GEV3(8)
8	East Madhya Pradesh	GEV3(5)
9	East Rajasthan	N2(6)
10	East Uttar Pradesh	PIII3(6)
11	Gangetic West Bengal	LN2(8) , LPIII3(8)
12	Gujarat	GEV3(5)
13	Haryana, Chandigarh & Delhi	LN2(9)

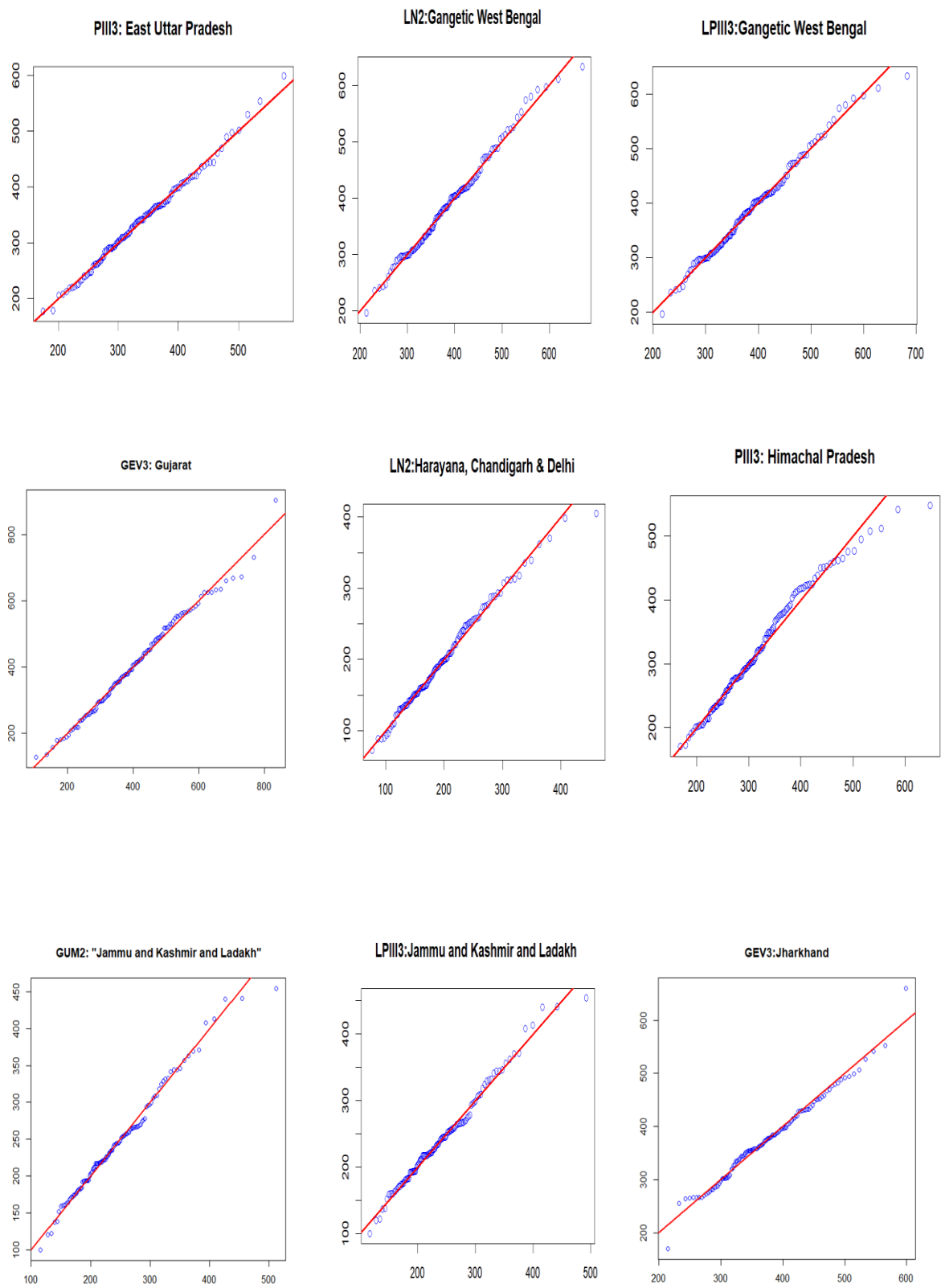
14	Himachal Pradesh	PIII3(7)
15	Jammu and Kashmir and Ladakh	LPIII3(8) , GUM2(8)
16	Jharkhand	PIII3(7) , GEV3(7)
17	Kerala	PIII3(6)
18	Konkan and Goa	N2(6) , GEV3(6)
19	Lakshadweep	GEV3(4)
20	Madhya Maharashtra	GEV3(4)
21	Marathwada	GEV3(4)
22	Nagaland, Manipur, Mizoram and Tripura	GUM2(5)
23	North Interior Karnataka	LN2(5)
24	Orissa	PIII3(9) , GEV3(9)
25	Punjab	LN2(8)
26	Rayalaseema	LN2(7) , GEV3(7)
27	Saurashtra, Kutch and Diu	GUM2(5)
28	South Interior Karnataka	GUM2(6)
29	Sub-Himalayan West Bengal and Sikkim	GEV3(4)
30	Tamil Nadu and Pondicherry	GEV3(5)
31	Telangana	LN2(7)
32	Uttaranchal	GEV3(7) , N2(7)
33	Vidarbha	LN2(7)
34	West Madhya Pradesh	LPIII3(7)
35	West Rajasthan	GEV3(5)
36	West Uttar Pradesh	W2(4)

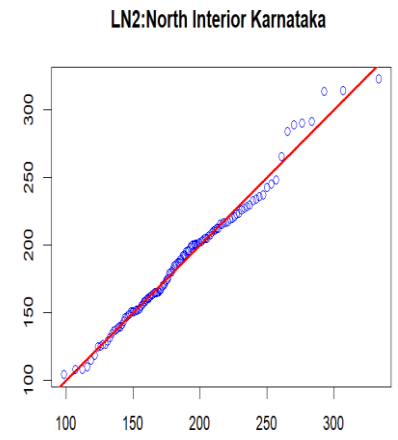
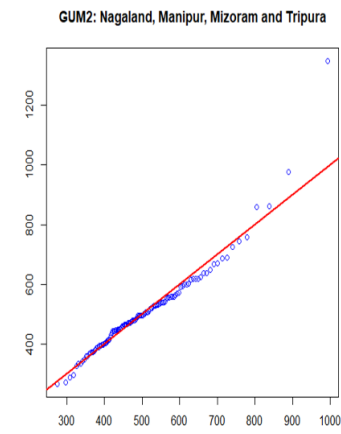
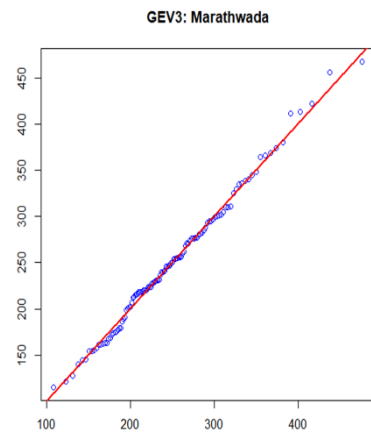
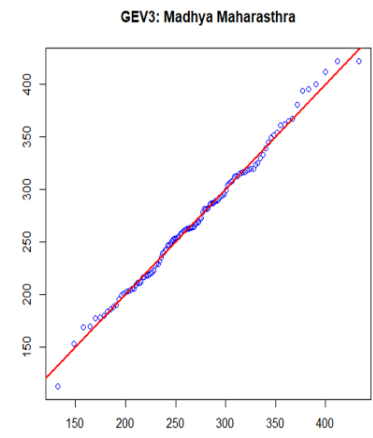
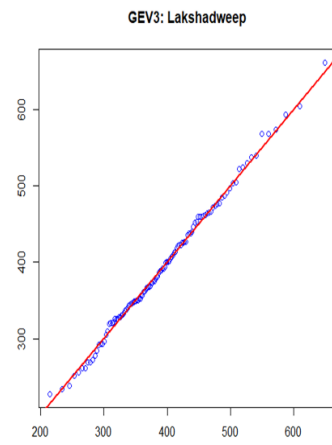
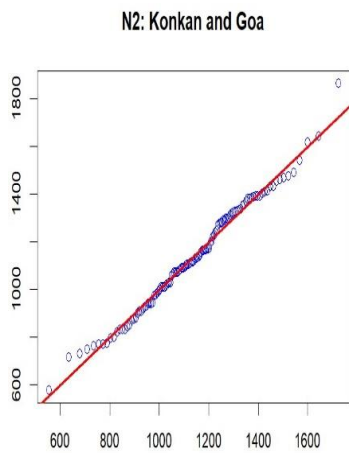
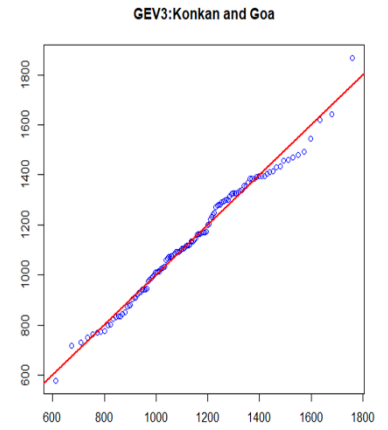
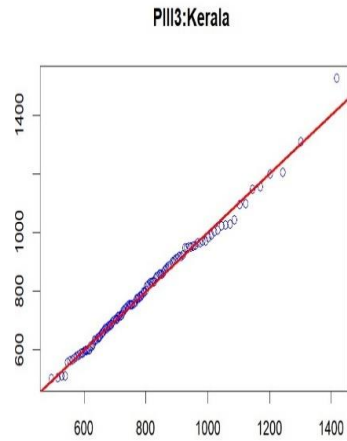
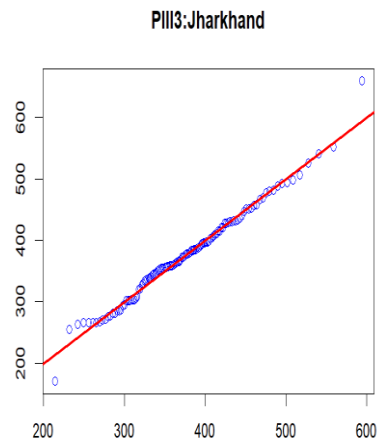
The Q-Q plots of each region was created using the best fit probability distributions. The test statistics which showed the smallest value using the four goodness of fit test such as K-S, AD, CVM, BIC was considered as the best fit and using the ranks, the probability distribution under given regions which showed lowest sum of the ranks was considered to be the best-fit probability distribution. The main purpose of the probability distribution fitting here is to most accurately represent low probability , extreme events on the extreme right tail. By using a Q-Q plot, the level of fit on the extreme right tail can be studied. We have used the best-fit probability distributions for each region and using their parameters values, estimated using the Maximum Likelihood Estimator (MLE) fitted the Q-Q plot for each region. Below we have shown Q-Q plots for each of the 36 regions using their best-fitted probability distribution. For e.g. the 2nd is of Arunachal Pradesh, whose data is best-fitted by PIII3 distribution , the Q-Q plot for the same has been plotted below. Bihar whose data is best fitted by GEV3 distribution; the Q-Q plot has been plotted below. Similarly, we have listed below Q-Q plots for all 36 regions. Of all the distribution with the lowest ranking, the GEV3 and PIII3 distributions best suit all the regions with mean maximum monthly rainfall above 400mm, with the exception of region number 22 ,which is best fitted by GUM2 distribution.

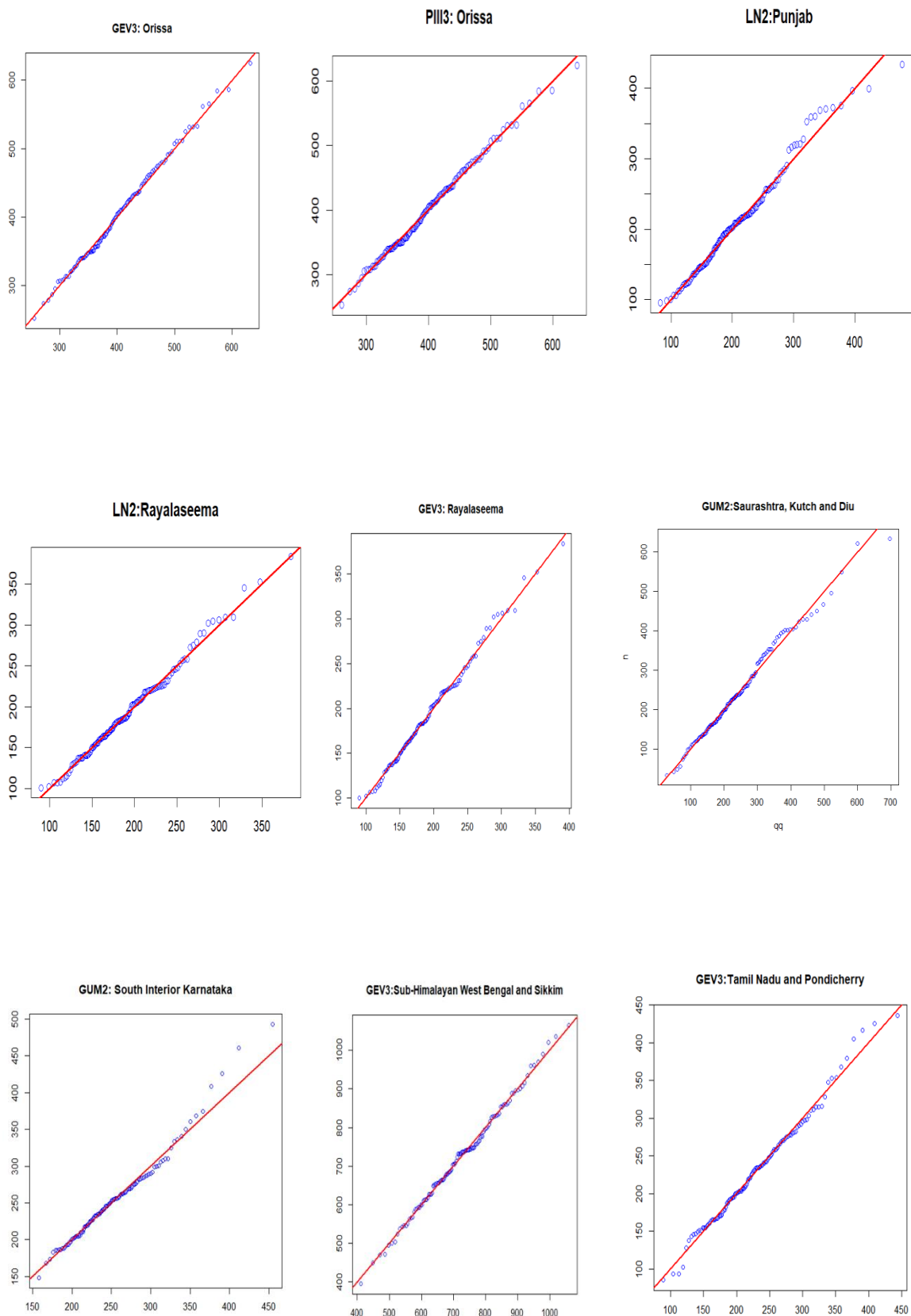
Figure: - 2

Q-Q plots of best-fitted Probability distribution of 36 regions

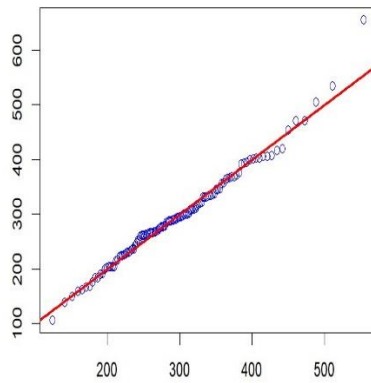




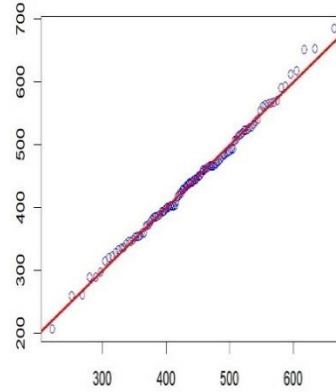




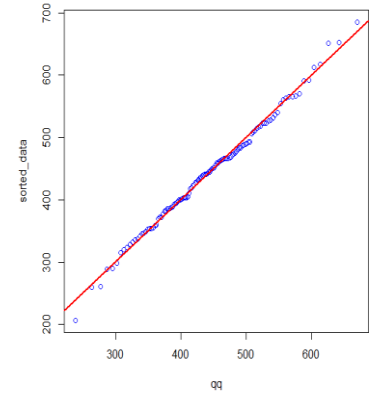
PII3:Telangana



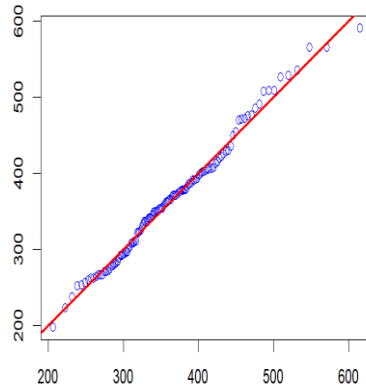
N2:Uttaranchal



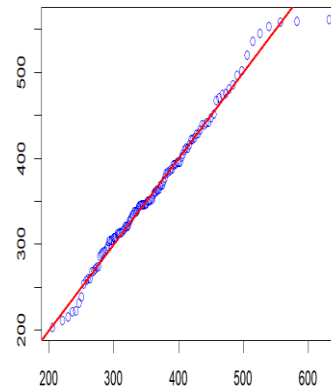
GEV3:Uttaranchal



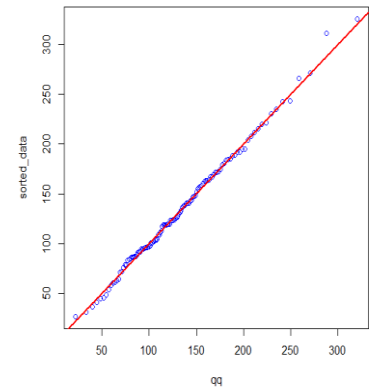
LN2:Vidarbha



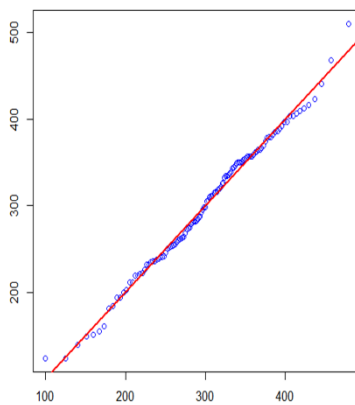
LPIII3:West Madhya Pradesh



GEV3:West Rajasthan



W2: West Uttar Pradesh



Using the observed and estimated values, QQ plots for each division are produced for the models with the best match. The 36 division's QQ plots are shown in the figure 2 above. The degree to which our selected model matches the gathered data is evident from the figure above.

Sub-Himalayan, Marathwada, Madhya Maharashtra, Lakshadweep, West Bengal and Sikkim, which yielded a GEV3 in all the four tests, is nearly a straight line indicating that the hypothesized distribution fits the data well. This was not always the case in Chattisgarh, where the GEV3 only showed a good match in two tests. Based on the ranking score Orissa's two best suited models are GEV3 and PIII3. The two distributions offered an excellent match to the data. The PIII3 distribution was shown to be the best fit in Arunachal Pradesh across all the four tests. However, the data in the middle and upper tails were poorly fitted. Similarly, W2, was shown to be the best match distribution across all the tests in West Pradesh, however the values are poorly fitted in both the tails.

In certain cases, such as those in Nagaland, South Interior Karnataka, Manipur, Mizoram, and Tripura, Costal Andhra Pradesh, Kerala, South Interior Karnataka and Telangana, the fitted distributions frequently underestimated the extreme rainfall values. These cases included large deviations in the upper tails. There were other instances when severe rainfall levels were overstated, meaning that the real observed values fell below the predicted line. These examples include the under Andaman and Nicobar Island, Punjab, Himachal Pradesh, Jammu and Kashmir and Ladakh, Haryana, Chandigarh and Delhi. The bottom tail has either no deviations at all or very minor ones.

There are other regions where the right tails appear to be fitted accurately by the presumed fitting distribution, including Chattisgarh, Lakshadweep, Sub-Himalayan West Bengal and Sikkim, Orissa and Uttaranchal. The data in regions like Chattisgarh and Sub-Himalayan West Bengal and Sikkim, Uttaranchal, Lakshadweep is best-fitted by the GEV3 distribution in the right tails. The distribution that best-fits the right tails data in Orissa were the GEV3 and PIII3 distributions. In the region of Costal Andhra Pradesh, LPIII3 distribution best fits the data in the right tails. LN2 best fits the right tails data in Haryana, Delhi and Chandigarh.

4.1.2. Return Period Results

According to the best fit probability model, we have then predicted return periods for 5 years, 10 years, 25years, 50 years ,100 years and 200 years. Table 6 shows the region names, their best fitted probability distribution along with the rainfall amounts for above specified return periods.

For e.g.: - for the Arunachal Pradesh region which is best fitted by PIII3 distribution, the rainfall amounts of 5, 10, 25, 50, 100, 200-year return periods are 1059.879mm, 71292.787mm, 1585.212mm, 1799.186mm, 2008.893mm, 2215.374mm respectively. Similarly, we have shown for each of the regions, the best-fit probability distribution and the rainfall amounts for above specified return periods.

In the north-eastern region, for example in Assam and Meghalaya, Nagaland, Manipur, Mizoram and Tripura receive very heavy rainfall. Landslides and floods commonly occur in these regions during the monsoon season, resulting in fatalities and property destruction. Flash floods are most commonly caused by exceptionally higher rainfall. Heavy rainfall in hilly region such as Himachal Pradesh, Jammu and Kashmir, Nagaland, Manipur, Mizoram and Tripura etc. even lead to soil erosion, glacier lake outburst floods (GLOFs) impacting infrastructure and livelihoods. The information of return periods can help guide policy makers and planners. They can make their decisions for a given span of time using the expected maximum yearly rainfall as determined by the return period.

Table 6. Regions, their best-fit Probability Distributions and Return Periods

S.NO.	Region Name	Probability Distribution	5 Years	10 Years	25 Years	50 Years	100 Years	200 Years
1	Andaman and Nicobar Island	PIII3	703.87	785.3214	885.0945	956.8723	1026.466	1094.405
		GEV3	697.2203	778.4566	884.6418	966.0546	1049.1595	1134.2978
2	Arunachal Pradesh	PIII3	1059.879	1292.787	1585.212	1799.186	2008.893	2215.374
3	Assam and Meghalaya	GUM2	664.218	732.24	818.186	881.9457	945.2346	1008.2925
		LPIII3	660.3528	716.8639	783.621	830.6929	875.9246	919.896
		GEV3	661.6871	720.4328	789.4927	837.2448	881.9015	923.8346
4	Bihar	GEV3	436.9203	473.2242	510.4321	532.9107	551.6708	567.4016
5	Chattisgarh	PIII3	503.3443	541.1528	583.5407	612.0761	638.5257	663.3878
		GEV3	505.862	539.7657	573.9178	594.1995	610.8841	624.6757
6	Coastal Andhra Pradesh	LPIII3	306.8088	346.7063	398.1949	437.427	477.4729	518.6636
		GEV3	306.2493	346.2423	398.1962	437.792	478.0058	518.9958
7	Coastal Karnataka	PV3	1368.756	1516.05	1701.093	1838.95	1977.305	2117.447
		GEV3	1369.819	1503.293	1660.245	1768.8	1870.341	1965.711
8	East Madhya Pradesh	GEV3	515.4724	562.1718	611.2463	641.6365	667.5323	689.7005
		PIII3	513.1017	563.7454	621.332	660.535	697.1598	731.8224
9	East Rajasthan	N2	339.7031	371.5814	405.576	427.5367	447.2898	465.3679
		GEV3	342.1563	373.1865	403.4458	420.8457	434.7776	445.9883
10	East Uttar Pradesh	PIII3	399.5592	440.1626	486.6501	518.4667	548.3017	576.6287
		GEV3	400.1298	441.4875	488.2517	519.3777	547.5633	573.1955
11	Gangetic West Bengal	LPIII3	455.6232	504.1317	562.3789	604.0207	644.4511	684.1266
		GEV3	456.1759	502.603	555.994	592.1248	625.3038	655.9
12	Gujarat	PIII3	517.6612	597.8195	691.8301	757.3634	819.5937	879.3119
		GEV3	517.7204	593.0241	678.2927	735.126	786.6511	833.563
13	Haryana, Chandigarh & Delhi	LN2	251.9664	293.7827	346.0481	384.6574	423.0523	461.5405
		PIII3	253.2786	292.7914	339.7739	372.8603	404.4948	435.026
14		LPIII3	251.7157	296.8826	355.6979	400.8046	447.0367	494.7398

		GEV3	251.4737	290.6174	338.0316	371.7894	404.1502	435.2943
15	Himachal Pradesh	LN2	389.4943	440.0991	501.3275	545.3392	588.218	630.4088
		PIII3	390.5919	445.4348	511.7854	559.1013	604.7172	649.0427
15	Jammu and Kashmir and Ladakh	LPIII3	293.457	333.5948	383.4405	420.1255	456.5448	493.0256
		GUM2	294.2863	336.3895	389.587	429.0519	468.2254	507.256
		GEV3	293.5674	332.1261	378.6965	411.7599	443.379	473.7371
16	Jharkhand	PIII3	435.5606	472.752	514.741	543.1655	569.616	594.5641
		GEV3	437.7259	476.9358	520.6888	549.4344	575.1806	598.3403
17	Kerala	PIII3	933.2903	1037.3406	1162.3594	1251.0726	1336.3211	1418.9386
		GEV3	928.3435	1031.5681	1159.5313	1252.718	1343.7733	1433.0886
18	Konkan and Goa	N2	1329.916	1430.027	1536.783	1605.748	1667.78	1724.552
		GEV3	1333.087	1445.248	1564.355	1638.88	1702.94	1758.255
19	Lakshadweep	GEV3	462.238	507.1211	557.7896	591.4607	621.9102	649.5643
20	Madhya Maharashtra	GEV3	320.9465	350.543	382.0519	401.8164	418.8411	433.5724
21	Marathwada	GEV3	305.3155	344.2044	389.6443	420.8779	449.9404	477.0943
22	Nagaland, Manipur, Mizoram and Tripura							
		GUM2	599.0094	675.2942	771.6803	843.1851	914.1618	984.8795
		GEV3	599.4586	678.5352	780.3386	857.2499	934.7826	1013.2291
23	North Interior Karnataka	LN2	221.1686	245.5485	274.5148	295.0196	314.7684	333.9989
		PIII3	221.6884	246.4182	275.4136	295.6215	314.8082	333.2182
24	Orissa	PIII3	462.3076	502.0276	548.2499	580.2822	610.5787	639.5548
		GUM2	462.6434	509.3755	568.4216	612.2254	655.7057	699.0274
		GEV3	461.5873	500.8747	546.1803	576.9228	605.2184	631.371
25	Punjab	LN2	264.4931	307.3255	360.67	399.9582	438.9403	477.9374
		LPIII3	263.7776	309.3676	368.2753	413.1474	458.8961	505.8675
		GEV3	262.9344	307.3152	364.337	407.3325	450.6039	494.3133
26	Rayalaseema	LN2	235.6934	266.8802	304.6968	331.9299	358.4988	384.6742
		GEV3	235.0755	267.2594	306.8422	335.4463	363.2141	390.2749

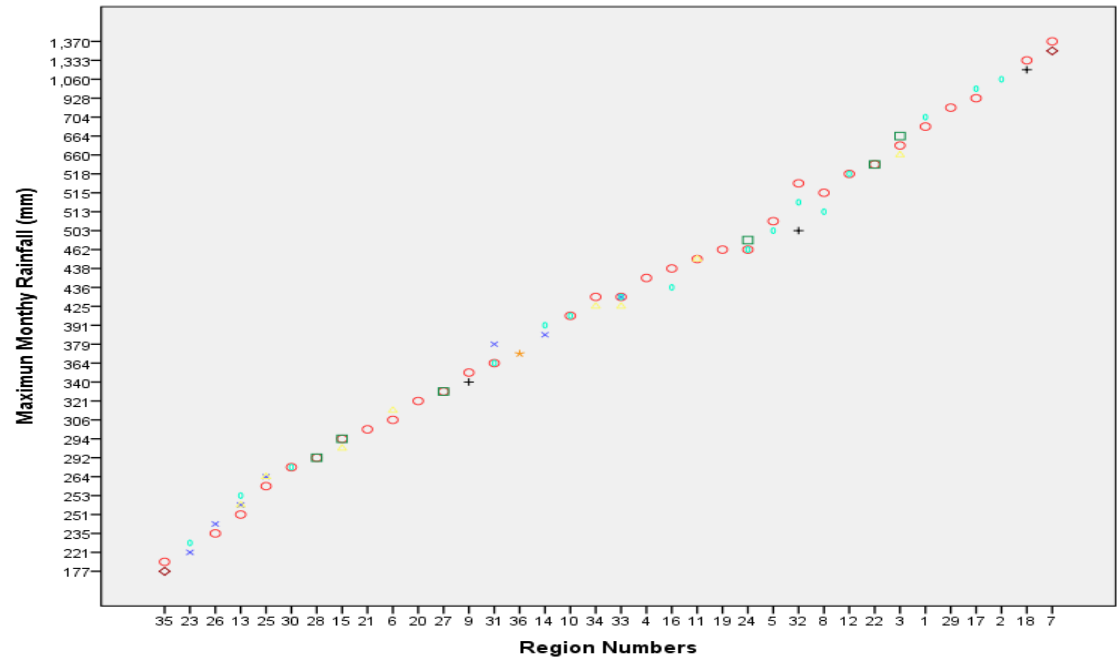
27	Saurashtra, Kutch and Diu	GUM2	329.7796	400.7709	490.4686	557.0114	623.0629	688.8734
		GEV3	329.6156	398.6584	484.5754	547.3758	608.9338	669.5049
28	South Interior Karnataka	GUM2	291.7945	323.3138	363.1384	392.6826	422.0087	451.2277
		GEV3	291.9809	325.0733	367.9966	400.6612	433.7935	467.5227
29	Sub- Himalayan West Bengal and Sikkim	GEV3	834.7621	896.4382	960.2359	999.1318	1031.8433	1059.4828
30	Tamil Nadu and Pondicherry	PIII3	284.9716	320.9809	362.3717	390.7866	417.4884	442.8864
		GEV3	284.9215	321.8647	364.2719	392.9184	419.1848	443.3701
31	Telangana	LN2	379.3645	440.7994	517.3118	573.6633	629.5756	685.5094
		PIII3	363.7305	407.3557	457.3859	491.6711	523.8498	554.4253
		GEV3	364.4857	411.3331	466.0907	503.7413	538.7846	571.5355
32	Uttaranchal	GEV3	516.7049	558.0732	601.2597	627.8297	650.3467	669.5174
		N2	503.4156	543.9251	590.7217	623.0719	653.6809	683.0102
		PIII3	513.6063	554.3426	599.0885	628.7175	655.8568	681.1026
33	Vidarbha	LN2	425.7348	467.5086	516.5812	550.988	583.8898	615.7202
		PIII3	426.45	468.4154	517.0893	550.7356	582.5041	612.8438
		LP III3	425.3939	468.5224	520.0387	556.7034	592.1802	626.8855
		GEV3	425.8917	467.8402	516.1696	548.9342	579.0679	606.8979
34	West Madhya Pradesh	LP III3	424.7994	469.1573	522.3099	560.2418	597.0203	633.0672
		GEV3	425.8496	467.1186	513.5876	544.3903	572.1866	597.3777
35	West Rajasthan	PV3	177.396	208.7223	246.4701	273.5741	300.0005	326.0507
		GEV3	177.7685	209.291	246.5654	272.4868	296.8456	319.8289
36	West Uttar Pradesh	W2	366.5209	396.8534	427.4937	446.4092	462.8722	477.5069

In table 6, 5-year,10-year, 25-year, 50-year ,100-year and 200-year rainfall return levels of all regions are illustrated. In the below given plots, the horizontal axis, ranging from 1 to 36, represents the region numbers which are given in Table 1. The vertical axis represents the maximum monthly rainfall (mm). Return periods for 5-year,10-year, 25-year, 50-year ,100-year and 200-year represented vertically from top to bottom. The return levels for each region are shown here corresponding to maximum monthly rainfall in mm. For choosing the best fit probability distribution for return period corresponding to each division we evaluated, the best-fit results of K-S, A-D, CVM, BIC according to ranks, given in table 4, and chose the minimum ranked distribution for each of K-S, A-D, CVM, BIC results corresponding to each division. For example, in Chattisgarh region, division number 5, the minimum ranked distribution for each of K-S, A-D, CVM, BIC results were PIII3, GEV3, PIII3, GEV3 respectively, therefore there were two different best fit distribution for return period at Chattisgarh region. In similar approach we calculated the best fit distribution for all the regions and concluded that GEV3 fitted 32 regions, PIII3 fitted 16 regions, LPIII3 fitted 8 regions, GUM2 fitted 6 regions, N2 fitted 3 regions, PV3 fitted 2 distributions and W2 fitted only 1 region.

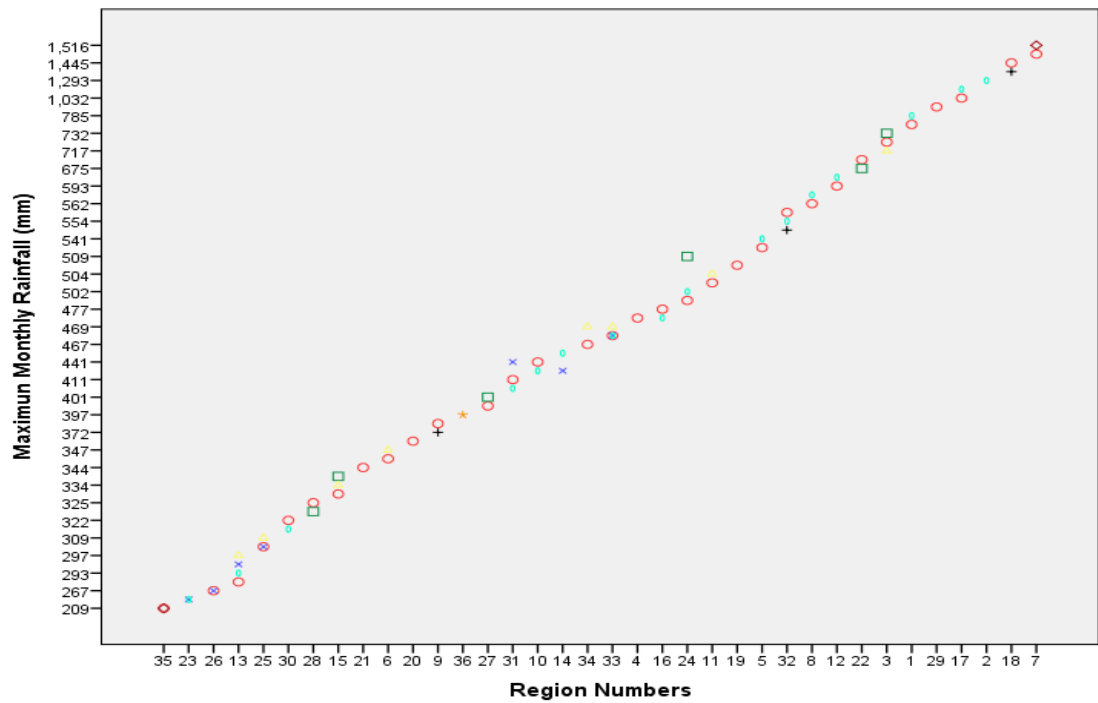
The return levels of the best-fit distributions for each region are shown in the Table 6, for each of 5, 10, 25, 50,100 and 200 years. As an example for Lakshadweep, region number 19, which is best-fitted by the GEV distribution, the rainfall amounts of 5, 10, 25-, 50-, 100- and 200-year return periods are 462.238 mm, 507.1211 mm, 557.7896 mm, 591.4607mm ,621.9102mm ,649.5643mm respectively. Below are the plots for each region corresponding to its return period and fitted distribution.

Legend corresponding to all the probability distribution used in the below graphs:

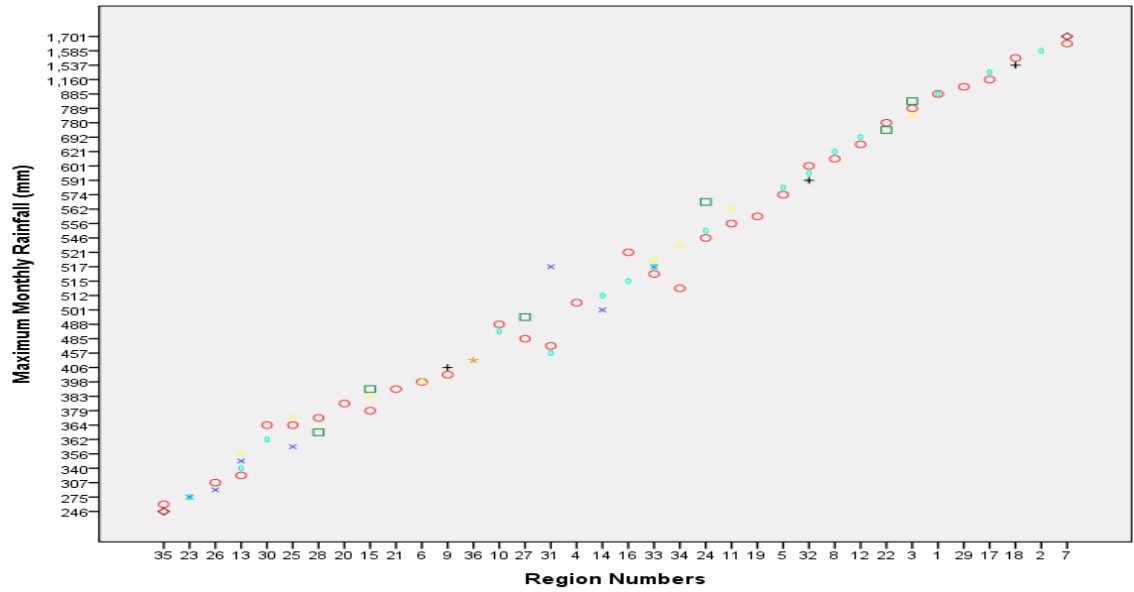
5 YEARS: Return Period



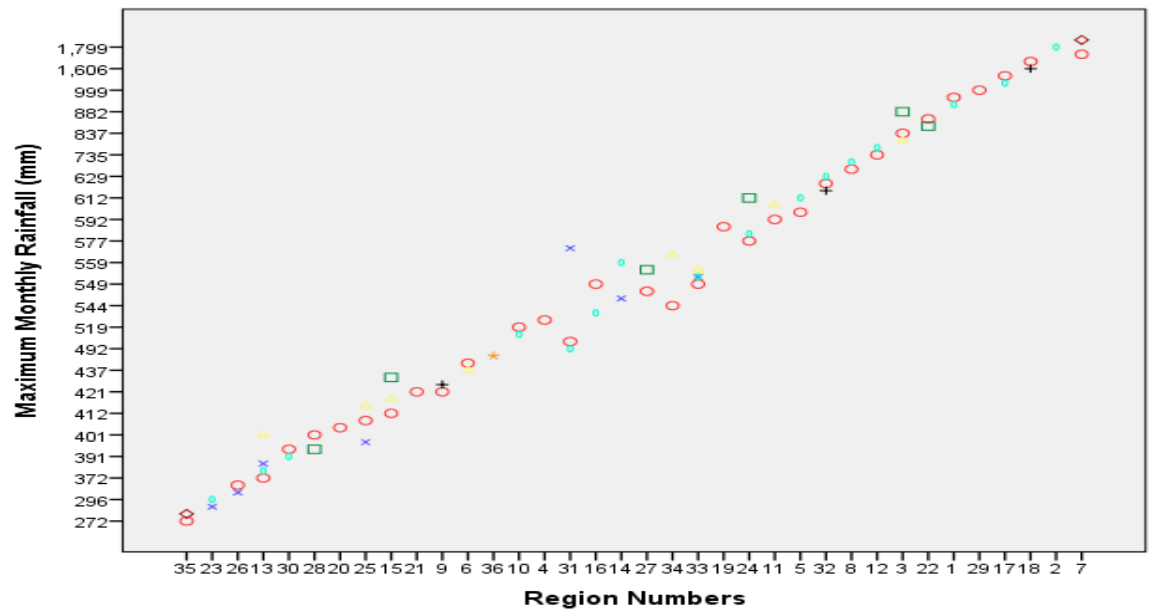
10 YEARS: Return Period



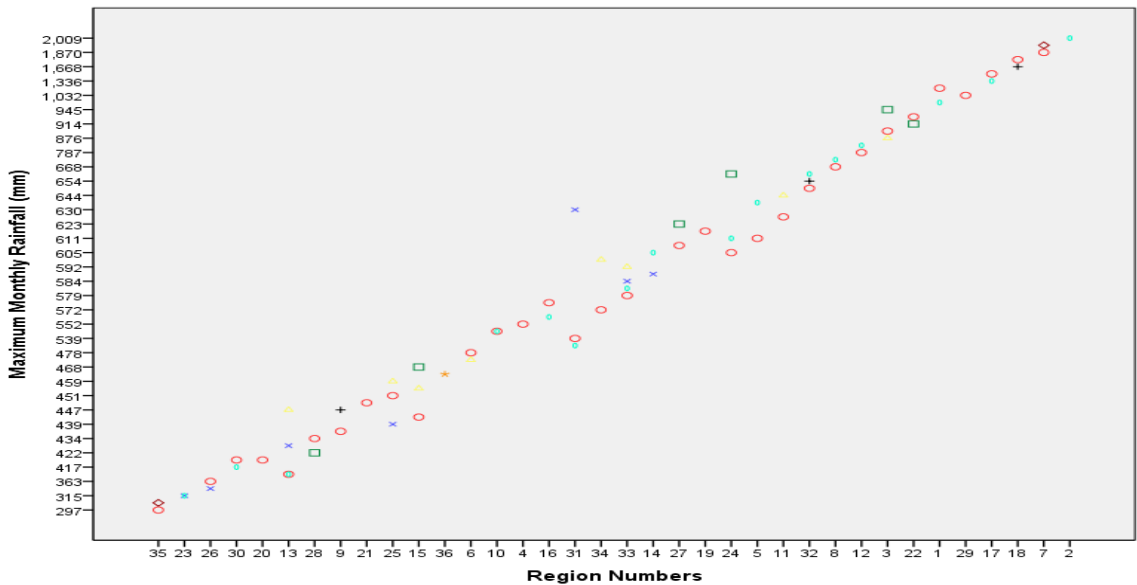
25 YEARS: Return Period



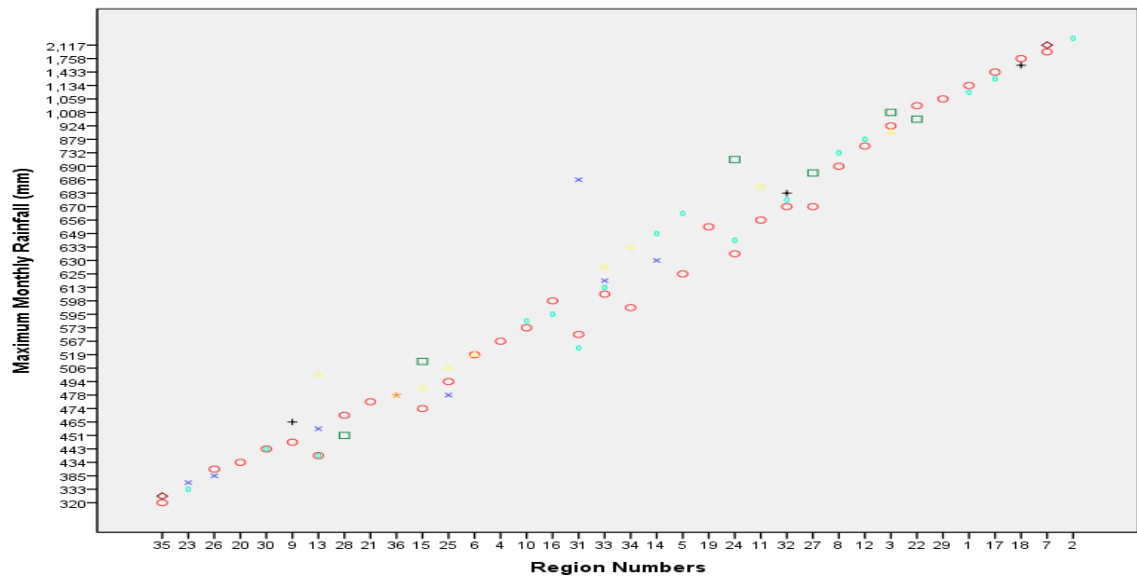
50 YEARS: Return Period



100 YEARS: Return Period



200 YEARS: Return Period



Probability Distribution



CONCLUSION:

The rainfall patterns vary across all locations. The site's surroundings and geographic location have a big impact on how the rainfall pattern varies. The most precipitation is measured in Arunachal Pradesh, while the least amount is measured in West Rajasthan, the findings differed greatly. We obtained the highest and lowest median, used standard deviation and skewness as a measure to divide our regions into categories. We have determined the best-fit probability distributions using lowest rank sum and using the four Goodness-of-fit tests such as K-S, AD, CVM and BIC to estimate the maximum annual rainfall for a given region. The maximum likelihood estimated was utilized to estimate the parameters for the distribution together with the graphical process called the Q-Q plot, the fits were determined using the K-S AD, CVM and BIC. Calculations were also made for the return period which is crucial in identifying the long-term concerns.

In our study, we determined areas with high and low mean Monthly rainfalls analyzed how the standard deviation values are distributed across various regions, used coefficient of skewness as a relative measure of variability to examine how standard division and mean rainfall numbers are related to one another. We have used median to examine the distribution in various geographic areas. Divided the region's coefficient of skewness into 3 categories: - severely skewed, moderately skewed and approximately symmetric.

For most of the regions the testing statistics yielded different best-fit distribution types. Thus, a scoring system was used to determine the best fit distribution for each region. The best fit result of each region was taken as a distribution with the lowest sum of rank scores from each test statistics. GEV3, PIII3 and LPIII3 showed the larger number of best-fit results compared to other distributions. In the best score result 47% was best-fitted by GEV3 distribution, 20% was best fitted by each of the PIII3 and LN2 distributions and 11% were best fitted by each of LPIII3 and GUM2 distributions. This aligns well in literature. The GEV3 was the most common best fit in research by Nguyen et al. 2016 for the extreme rainfall frequency analysis. Also, the GEV3 was the most common best fits for extreme rainfall series by Zalina et al. 2002 in Malaysia, by Ghosh et al. 2016 for monthly rainfall in Bangladesh. PIII3 was the most common best-fits by Hanson et al. 2008 on daily rainfall in US, LN2 was best fitted by Douka et al. 2018 for extreme rainfall analysis, by Ng et al. 2018 for precipitation of tropical climate. LPIII3 was the most common best-fits by Farooq et al. 2018 for the flood frequency analysis, Yuan et al. 2018 for annual maximum hourly precipitation.

The more practical result of this study was the return period calculation for all the regions and all distributions. The presentation of all distributions gave a clear inspection of the range of projected return period. Here 5-year, 10-year, 25-year, 50-year, 100-year and 200-year return periods of rainfall levels were illustrated.

Better models of the danger and damage from floods and periods of heavy rainfall may be created using the findings of this study. These predictions can aid decision makers in developing programs that will ultimately save lives and property. Additionally, understanding the trend of really heavy rainfall would be helpful in many fields, including agriculture, construction, the environment and building architecture.

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